

Local-Unitary Rigidity and Quantitative Rounding for Stabilizer AME States

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July 2026

Abstract

Let $q = p^e$ and $m \geq 2$. We prove that every product unitary mapping one stabilizer $\text{AME}(2m, q)$ state to another is Clifford on each party. The key support-counting statement is that the stabilizers contained in any $(m + 1)$ -party set project bijectively onto the complete Weyl-label space at each retained party. The corresponding reduced operator therefore determines every local Weyl axis. It follows that local-unitary and local-Clifford equivalence coincide for these states. Equivalently, every transversal conversion between the associated $[[2m - 1, 1, m]]_q$ stabilizer encoders is Clifford on every physical and logical factor. The same support data give a complete classification by the transition maps between local Weyl-label spaces.

We also prove a quantitative result for approximate product symmetries. Three-region cleaning and a finite Weyl–Fourier argument place every local factor within normalized Hilbert–Schmidt distance 8ε of a Clifford. A uniform gap between distinct stabilizer states then selects an exact symmetry and yields an explicit decomposition radius of order

$$\min\{p^{-1}, q^{-1/2}, n^{-1/2}\}, \quad n = 2m,$$

with collective residual generator norm at most $\pi\sqrt{q}\varepsilon$. At a larger, dimension-dependent radius the rounded symplectic maps already preserve all minimum-support transitions; the remaining difficulty is to control the stabilizer character, or equivalently the required product-Pauli correction.

Index Terms—Absolutely maximally entangled states, local Clifford group, local-unitary equivalence, quantitative rigidity, quantum MDS codes, stabilizer cleaning, transversal gates, Weyl operators.

1 Introduction

An absolutely maximally entangled state has maximally mixed reductions on every set containing at most half of its parties. Such states are also called perfect tensors and, after one party is designated as an input, give quantum maximum-distance-separable encoders. They therefore sit at the intersection of multipartite entanglement, stabilizer codes, transversal gates, and tensor-network constructions.

This paper asks how rigid their local coordinates are. With the party labels fixed, two states are *local-unitary (LU) equivalent* if a tensor product of one-party unitaries maps one state ray to the other. We say *LU up to party relabelling* when a party permutation is also allowed. The analogous relations are *local-Clifford (LC) equivalence* and LC equivalence up to party relabelling, with every one-party factor required to normalize the chosen Weyl, or Pauli, system. The theorem below uses the latter convention; setting the permutation to the identity gives the fixed-label statement. Local-Clifford equivalence is discrete and can be studied through finite symplectic linear algebra; local-unitary equivalence is a priori

continuous and much larger. The two relations do not coincide for general stabilizer states [JCWY10].

Maximal entanglement removes that difference.

Theorem 1.1 (Local-unitary rigidity). Let $q = p^e$, let $m \geq 2$, and let $|\psi\rangle, |\phi\rangle$ be stabilizer $\text{AME}(2m, q)$ states for the standard additive Weyl system. Suppose that, for a party permutation π , single-qudit unitaries $U_1, \dots, U_{2m}$, and a phase θ ,

$$(U_1 \otimes \dots \otimes U_{2m})\pi|\psi\rangle = e^{i\theta}|\phi\rangle.$$

Then every U_i normalizes the Weyl system. Thus the displayed equivalence is local Clifford.

The theorem converts a continuous equivalence problem into a finite one. It is valid for every prime power, for arbitrary additive stabilizers, and does not assume CSS form, equal stabilizer phases, classical linearity over $\mathbb{F}_q$, or a particular construction of the AME state. The restriction $m \geq 2$ is sharp: a Bell pair has the continuous symmetry $U \otimes \bar{U}$.

The proof uses one elementary support calculation. On any $(m+1)$ -party set, the stabilizer labels supported there form a group of order q^2 . Projection onto the two-dimensional Weyl-label space of any retained party is bijective. Consequently the reduced operator contains every local Weyl axis exactly once in a matched diagonal decomposition. The uniqueness of that decomposition forces each local unitary to permute the Weyl axes and hence to be Clifford. In ordinary language,

$$\text{AME support counting} \implies \text{all local Weyl axes appear} \implies \text{every local factor is Clifford.} \quad (1.1)$$

The operational code consequence follows by treating one party as the logical input of the associated quantum MDS encoder. Throughout, *transversal* means a tensor product of one-site physical operators with the physical coordinates fixed.

Corollary 1.2 (Factorwise transversal rigidity). Let $V_\psi, V_\phi : \mathbb{C}^q \rightarrow (\mathbb{C}^q)^{\otimes(2m-1)}$ be encoders whose normalized Choi states are stabilizer $\text{AME}(2m, q)$ states. If

$$(U_1 \otimes \dots \otimes U_{2m-1})V_\psi = V_\phi L,$$

then L and every U_i are Clifford. In particular, no code in this family admits a transversal non-Clifford logical unitary.

Quantitative question

Exact rigidity does not by itself say what happens when a product unitary only approximately preserves the state. For $U = \bigotimes_i U_i$, define its state-vector defect by

$$\varepsilon(U) = \min_{|z|=1} \|U|\psi\rangle - z|\psi\rangle\|.$$

The direct marginal proof is poorly conditioned for this problem: the nonidentity part of one minimum-support marginal has exponentially small Hilbert–Schmidt norm. We instead regard each party in turn as the logical leg of the associated $[[2m-1, 1, m]]_q$ code. Its physical parties can be partitioned into three correctable regions. A leakage-aware cleaning argument tests the induced logical action by nested commutators, and a finite Fourier argument rounds that action to a Clifford.

The exact constants are useful for applications, so we state them once in the introduction. Their scaling and proof mechanism are the main points. Write $G(\psi)$ for the group of product unitaries that preserve the ray of $|\psi\rangle$.

Theorem 1.3 (Quantitative rounding). Let $|\psi\rangle$ be a stabilizer AME(n, q) state with $n = 2m$, $m \geq 2$, and let $U = \bigotimes_i U_i$. Put

$$\begin{aligned}\tau_p &= \min \left\{ \frac{1}{3}, \frac{\sin(\pi/p)}{2\sqrt{2}} \right\}, & d_p &= (2 - 2p^{-1/2})^{1/2}, \\ R_{n,q}^{\text{clean}} &= \min \left\{ \frac{\tau_p}{8}, \frac{1}{4\sqrt{2q}}, \frac{1}{8\pi\sqrt{n}}, \frac{d_p}{1 + 4\pi\sqrt{n}} \right\}.\end{aligned}\tag{1.2}$$

If $\varepsilon(U) < R_{n,q}^{\text{clean}}$, then, up to phase,

$$U = g \bigotimes_{i=1}^n e^{ih_i}, \quad g \in G(\psi), \quad \left(\sum_i \|h_i\|_F^2 \right)^{1/2} \leq \pi\sqrt{q}\varepsilon(U),\tag{1.3}$$

where the h_i are traceless Hermitian with spectral spread $\lambda_{\max}(h_i) - \lambda_{\min}(h_i)$ at most π . Before the exact symmetry g is selected, each U_i is within normalized Hilbert–Schmidt distance $8\varepsilon(U)$ of a one-qudit Clifford. Uniformly over prime powers and existing AME states,

$$R_{n,q}^{\text{clean}} = \Theta\left(\min\{p^{-1}, q^{-1/2}, n^{-1/2}\}\right).$$

The proof has three stages. First, quantitative cleaning places every nested logical Weyl commutator within 8ε of a scalar. Second, Fourier concentration on the finite Weyl group produces a nearby local Clifford at each party. Third, the quantized overlap of stabilizer states separates distinct product-Clifford branches by the uniform gap d_p . Once the correct exact branch is known, the AME second moment gives the collective residual estimate in (1.3). The radius is a certified lower bound, not an optimality claim.

An exact product-unitary intertwiner between two states transfers this result without changing the constants (Corollary 5.6). Without such a base intertwiner, Proposition A.21 detects the correct orbit only at the weaker scale $q^{-(m+1)/2}$, because that route must read the small nonidentity signal of one marginal.

Transition maps and the remaining obstruction

The support bijections retain how the local Weyl frames are related. For an $(m+1)$ -party support A , projection at party i gives a bijection $p_i : L_\psi(A) \rightarrow \mathbb{F}_q^2$. The maps

$$M_A(i, j) = p_j p_i^{-1}$$

are the usual operator-pushing maps between the local Weyl-label spaces. Their compositions around closed paths are holonomies.

Theorem 1.4 (Classification by minimum-support transition maps). After a fixed party relabelling, two stabilizer AME($2m, q$) states are LU equivalent exactly when their systems of minimum-support transition maps are equivalent by local trace-symplectic changes of Weyl frame. For one state, if $\text{Aut}_{\text{tr}}(\psi)$ is the group of compatible local symplectic automorphisms of the transition system and Γ_ψ its fixed-party projective product-unitary symmetry group, then

$$1 \longrightarrow L_\psi \longrightarrow \Gamma_\psi \longrightarrow \text{Aut}_{\text{tr}}(\psi) \longrightarrow 1$$

is exact. When q is prime, $\text{Aut}_{\text{tr}}(\psi)$ is the centralizer in $\text{SL}_2(q)$ of the transition holonomies after base transports are chosen.

At the dimension-dependent radius

$$\min \left\{ \frac{\tau_p}{8}, \frac{2 \sin(\pi/p)}{4 + 64\sqrt{q}} \right\}, \quad (1.4)$$

the symplectic maps recovered by cleaning already preserve every transition map exactly. This does not yet identify the exact product symmetry. Localized commutators cancel Weyl-translation phases, so they determine the stabilizer label space but not its character. The resulting product-Pauli correction need not be close to the rounded product Clifford. We refer to this standard linear-versus-affine distinction as the *stabilizer-character obstruction*. It is the missing ingredient in a party-count-independent global entry radius.

Relation to prior work and contribution

The exact three-cleanable-region Clifford obstruction is due to Pastawski and Yoshida [PY15, Lemma 5], building on stabilizer cleaning [BT09, Lemma 1] and the localization method behind transversal-gate hierarchy bounds [BK13, JOKY18]. We use only the exact cleaning and localization consequences and prove the leakage estimate, finite Weyl–Fourier rounding, and explicit AME composition here.

For exact LU rigidity, Rains recovered the three Pauli axes from a diagonal correlation tensor, and Van den Nest, Dehaene, and De Moor used that mechanism in a minimum-support criterion for qubit stabilizer states [Rai99, Theorem 13] [VdNDDM05, Theorem 1 and Corollary 1]. Englbrecht and Kraus classify all qubit stabilizer local symmetries [EK20]; graphical and algorithmic descriptions of qubit stabilizer LU equivalence are also available [CP24, BdJV24]. The present support argument extends the axis-recovery mechanism to arbitrary additive prime-power stabilizers and, because every minimum support contains the full local Weyl basis, requires no additional minimum-support hypothesis.

Tan proves the general AME–quantum-MDS orbit and symmetry–transversal correspondences [Tan26b, Theorems 3.5 and 3.6] and computes particular four-qutrit and five-qubit symmetry groups [Tan26b, Tan26a, Theorem 5.3]. Our exact conclusion is factorwise Cliffordness for every conversion in the stabilizer-AME family, together with the complete transition-map invariant. The new quantitative conclusion is the explicit cleaning-based rounding theorem and its separation of symplectic compatibility from the remaining stabilizer character. No new qualitative transversal-hierarchy obstruction is claimed.

The stabilizer assumption is essential to the discrete transition system. General AME families can contain infinitely many LU classes; Ramadas and Lakshminarayan give an orthogonal-array criterion producing such families [RL25, Theorem 1 and Corollary 1]. Perfect-tensor operator pushing and the encoder interpretation are used explicitly in holographic codes [PYHP15, Eq. (2.3) and Section 5.7].

Organization

Section 2 fixes the additive stabilizer conventions and derives the AME support formula. Sections 3 and 4 prove exact rigidity, the transition-map classification, and the encoder corollary. Section 5 proves cleaning-based rounding, robust transition compatibility, and the stabilizer-character obstruction. Section 6 summarizes the scaling and open problem.

The appendices are independent of the headline proofs. Appendix A develops alternative stability mechanisms and weaker comparison radii. The final appendix states the formal and citation trust boundary.

2 Stabilizer AME support geometry

Let $q = p^e$ and fix the additive character $\chi(a) = \exp(2\pi i \operatorname{Tr}_{\mathbb{F}_q/\mathbb{F}_p}(a)/p)$. On the standard basis of $\mathbb{C}^q$, indexed by $\mathbb{F}_q$, put

$$X(a)|x\rangle = |x+a\rangle, \quad Z(b)|x\rangle = \chi(bx)|x\rangle.$$

For $v = (a, b) \in \mathbb{F}_q^2$, write $W_v = X(a)Z(b)$, with any fixed consistent phase convention when only its projective axis is used. The q^2 single-party operators W_v form an orthogonal basis of the single-party Hilbert–Schmidt space. A unitary is Clifford when conjugation permutes their projective axes $\mathbb{C}W_v$.

The label-space commutator is measured by the *trace-symplectic pairing*

$$[v, w] = \operatorname{Tr}_{\mathbb{F}_q/\mathbb{F}_p}(ab' - ba') \in \mathbb{F}_p, \quad v = (a, b), \quad w = (a', b').$$

After choosing dual $\mathbb{F}_p$ -bases of $\mathbb{F}_q$, this Weyl system is the Pauli system of e p -level factors. Its projective Clifford normalizer has the standard exact sequence

$$1 \longrightarrow \mathbb{F}_p^{2e} \longrightarrow \operatorname{PCliff}(q) \longrightarrow \operatorname{Sp}_{2e}(\mathbb{F}_p) \longrightarrow 1, \quad (2.1)$$

including $p = 2$: the kernel is the projective Pauli group, and every trace-symplectic $\mathbb{F}_p$ -linear label map has a Clifford lift [DDM03, HDDM05].

For n parties, a pure stabilizer state is fixed by an abelian Pauli group $\mathcal{S}$ of order q^n . Its projective label group $L \leq (\mathbb{F}_q^2)^n$, viewed additively over $\mathbb{F}_p$, is maximal isotropic for the trace-symplectic form. This is the standard arbitrary-additive prime-power stabilizer correspondence [KKKS06, Theorems 13 and 15]. For each $\ell \in L$, write the stabilizer lift as $\widetilde{W}_\ell = \alpha_\ell W_\ell$, where $\alpha_\ell \neq 0$. These phases encode the joint eigenvalue assignment on L , relative to the fixed Weyl convention; we call that assignment the *stabilizer character*. Then

$$|\psi\rangle\langle\psi| = q^{-n} \sum_{\ell \in L} \widetilde{W}_\ell. \quad (2.2)$$

If A is a party set, put

$$L(A) = \{\ell \in L : \operatorname{supp}(\ell) \subseteq A\}.$$

Weyl orthogonality and partial trace give

$$\rho_A^\psi = q^{-|A|} \sum_{\ell \in L(A)} \widetilde{W}_\ell. \quad (2.3)$$

In particular, ρ_A^ψ is maximally mixed exactly when $L(A) = \{0\}$. Thus a stabilizer state on $2m$ parties is AME exactly when it has no nonidentity stabilizer label supported on at most m parties.

Equations (2.2)–(2.3) are the only state-specific input to the exact rigidity theorem. The same support geometry also identifies any chosen party as the logical leg of a one-qudit stabilizer encoder on the remaining $2m - 1$ parties. The resulting code has parameters $[[2m - 1, 1, m]]_q$: erasure of at most $m - 1$ physical parties is correctable because the complementary reductions of the AME state are maximally mixed. We use this encoder interpretation twice, first for the exact transversal consequence and then for quantitative cleaning.

3 Complete Weyl-basis marginals and LU rigidity

We first isolate the linear-algebra statement that recovers a diagonal tensor’s coordinate axes. It is the complete-Weyl-basis counterpart of the three-Pauli argument in [Rai99, VdNDDM05].

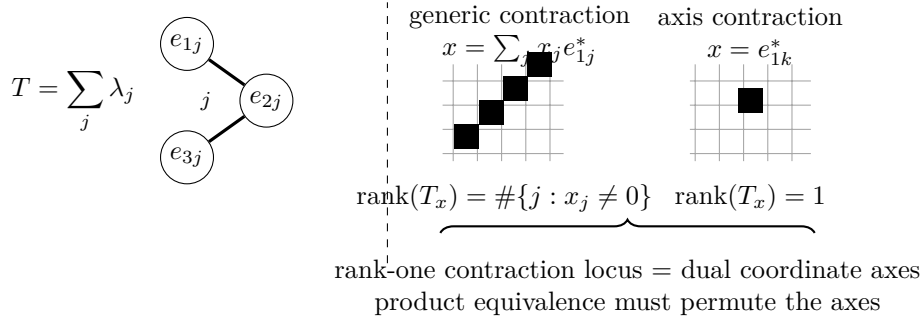


Figure 1: A diagonal tensor with all coefficients nonzero exposes its own axes. Contracting one leg against x and flattening the remainder gives rank equal to the number of nonzero coordinates of x , so the rank-one contraction locus is exactly the set of dual coordinate axes. For an $(m + 1)$ -party AME marginal in local Hilbert–Schmidt spaces, the recovered axes are the local Weyl axes, and a product conjugation must permute them.

The statement itself is classical: it is Jennrich’s uniqueness theorem for trilinear decompositions, published in [Har70, §V] and there credited to Jennrich, whose conclusion that the factor matrices agree up to one permutation and compensating diagonal scalings is the monomial property below. It is the full-column-rank case of Kruskal’s condition [Kru77], extended to arbitrarily many factors in [SB00]; see [KB09, §3.2] for the attribution and for a survey of the uniqueness theory. We give a short self-contained proof because the contraction argument is the one used in Proposition 3.2, and claim no novelty for either the statement or the proof.

Lemma 3.1 (Diagonal-tensor axes). Let $r \geq 3$, let $E_1, \dots, E_r$ be N -dimensional complex inner product spaces, and choose orthonormal bases $\{e_{ij} : 1 \leq j \leq N\}$. If

$$T = \sum_{j=1}^N \lambda_j e_{1j} \otimes \cdots \otimes e_{rj}, \quad \lambda_j \neq 0, \quad (3.1)$$

then the coordinate axes in every E_i are intrinsic to T . Consequently, a product of invertible linear maps carrying one tensor of the form (3.1) to another is monomial in the displayed bases.

Proof. For nonzero $x = \sum_j x_j e_{1j}^*$, contract the first factor against x . The result is

$$T_x = \sum_j \lambda_j x_j e_{2j} \otimes \cdots \otimes e_{rj}.$$

Flattening from E_2 to $E_3 \otimes \cdots \otimes E_r$ gives a matrix of rank $\#\{j : x_j \neq 0\}$. Thus T_x has rank one exactly when the ray of x is a dual coordinate axis. Product equivalence preserves this projective rank-one contraction locus. The inner products identify the recovered dual axes with the coordinate axes in the first factor. Apply the same argument in every factor. $\square$

To avoid repeatedly stating the matched-basis condition, call an operator on $r \geq 3$ qudits *Weyl-diagonal with complete local support* if there is a q^2 -element set P , a distinguished element $0 \in P$, and bijections $f_i : P \rightarrow \mathbb{F}_q^2$ with $f_i(0) = 0$, for which

$$R = \sum_{v \in P} \lambda_v \bigotimes_{i=1}^r W_{f_i(v)}, \quad \lambda_v \neq 0. \quad (3.2)$$

The coefficients need not be equal. What matters is that the same q^2 -element index set runs bijectively through the complete local Weyl basis at every party. Neither P nor the bijections need be $\mathbb{F}_q$ -linear.

Proposition 3.2 (Complete Weyl-basis marginal criterion). Let R and R' be Weyl-diagonal operators with complete local support on the same $r \geq 3$ parties, possibly with different indexing sets, bijections, and coefficients. If

$$R' = (U_1 \otimes \cdots \otimes U_r) R (U_1 \otimes \cdots \otimes U_r)^\dagger,$$

then every U_i is Clifford.

Proof. View R and R' as tensors in the local Hilbert–Schmidt spaces. After replacing W_v by the orthonormal basis $q^{-1/2}W_v$ and absorbing the common normalization into the coefficients, both have the form in Lemma 3.1, with $N = q^2$. Local conjugation acts invertibly on each factor, so it must permute the Weyl axes. It fixes the identity operator and hence the identity axis. It therefore permutes the nonidentity projective Weyl operators, which is exactly the single-qudit Clifford condition. $\square$

Corollary 3.3 (Marginal-cover rigidity). Let $|\psi\rangle, |\phi\rangle$ be n -qudit states and suppose a product unitary carries $|\psi\rangle$ to $|\phi\rangle$ up to phase, any party permutation having first been absorbed by relabelling $|\phi\rangle$, so that the two states are indexed by the same parties. If every party belongs to a set S of at least three parties for which both reduced operators ρ_S^ψ and ρ_S^ϕ are Weyl-diagonal with complete local support, then every local factor is Clifford.

Proof. Partial trace gives a product-unitary equivalence of the two reduced operators on each chosen S . Apply Proposition 3.2; the chosen sets cover all parties. $\square$

Proposition 3.4 (Stabilizer-AME support theorem). Let $|\psi\rangle$ be a stabilizer AME($2m, q$) state, and let A be any $m + 1$ parties. Then

$$|L(A)| = q^2.$$

Every nonidentity label in $L(A)$ has support exactly A , and for each $i \in A$ the coordinate projection

$$p_i : L(A) \longrightarrow \mathbb{F}_q^2$$

is bijective.

Proof. The projective label group L has order q^{2m} . Restrict its labels to the $m - 1$ parties outside A . The target has order $q^{2(m-1)}$, and the kernel is $L(A)$; hence

$$|L(A)| \geq q^{2m} / q^{2(m-1)} = q^2. \quad (3.3)$$

Fix $i \in A$. A label in the kernel of p_i is supported on the m -set $A \setminus \{i\}$. The AME condition and (2.3) exclude every such nonidentity stabilizer, so p_i is injective. Its codomain has order q^2 , giving the reverse inequality in (3.3) and hence bijectivity. The same no-small-support argument shows that every nonidentity element of $L(A)$ has full support A . $\square$

Proposition 3.5 (A stabilizer-AME marginal fixes the local Weyl axes). Let $|\psi\rangle, |\phi\rangle$ be stabilizer AME($2m, q$) states. If a product unitary on an $(m + 1)$ -set A carries ρ_A^ψ to ρ_A^ϕ , then every local factor on A is Clifford.

Proof. By (2.3),

$$\rho_A^\psi = q^{-|A|} \sum_{\ell \in L_\psi(A)} \widetilde{W}_\ell.$$

Index this sum by $P = L_\psi(A)$. Proposition 3.4 says that every coordinate projection $P \rightarrow \mathbb{F}_q^2$ is bijective. Relative to the fixed Weyl convention, each stabilizer lift contributes a nonzero coefficient. Thus ρ_A^ψ is Weyl-diagonal with complete local support in the sense of (3.2), and the same holds for ρ_A^ϕ . Apply Proposition 3.2. $\square$

Proof of Theorem 1.1. Absorb the party permutation by relabelling $|\phi\rangle$. Partial trace gives a product-unitary equivalence $\rho_A^\psi \sim \rho_A^\phi$ for every $(m+1)$ -set A . Given a party i , choose such a set containing it and apply Proposition 3.5. Thus every U_i is Clifford. $\square$

4 Minimum-support transition maps and encoder consequences

The same support calculation recovers standard quantum-MDS structure. If $|A| \leq m$, the AME condition gives $L(A) = \{0\}$. If $|A| > m$, then $\rho_{A^c}^\psi$ is maximally mixed, so purity of the global state gives $\text{Tr}((\rho_A^\psi)^2) = q^{-(2m-|A|)}$. On the other hand, (2.3) and Weyl orthogonality give $\text{Tr}((\rho_A^\psi)^2) = q^{-|A|}|L(A)|$. Hence, for every party set A ,

$$|L(A)| = q^{2\max(|A|-m, 0)}. \quad (4.1)$$

Möbius inversion is the stabilizer-state specialization of the quantum-MDS Shor–Laflamme distribution of Huber and Grassl [HG20, Theorems 8 and 9]; in particular the number of weight- $m+1$ projective stabilizers is $\binom{2m}{m+1}(q^2-1)$. We use one further consequence. If $|A| \geq m+2$ and $i \neq j$ lie in A , then the dimensions in (4.1), together with

$$L(A \setminus \{i\}) \cap L(A \setminus \{j\}) = L(A \setminus \{i, j\}),$$

give

$$L(A) = L(A \setminus \{i\}) + L(A \setminus \{j\}). \quad (4.2)$$

Descending induction shows that the $(m+1)$ -supported subgroups span the full label group. This is the additive quantum-MDS form of minimum-weight generation, not a separate weight-enumerator claim.

We now prove the transition-map classification stated in the introduction. Regard each local label group

$$P_i = \mathbb{F}_q^2$$

as a $2e$ -dimensional trace-symplectic space over $\mathbb{F}_p$. For every $(m+1)$ -set A and $i, j \in A$, the support theorem defines the operator-pushing transition

$$M_A(i, j) = p_j p_i^{-1} : P_i \longrightarrow P_j.$$

For two states ψ, ϕ on the same labelled parties, a *symplectic equivalence of the transition systems* is a tuple $F_i \in \text{Sp}(P_i)$ satisfying

$$F_j M_A^\psi(i, j) = M_A^\phi(i, j) F_i \quad \text{for every } A \text{ and } i, j \in A. \quad (4.3)$$

Here $\text{Sp}(P_i) = \text{Sp}_{2e}(\mathbb{F}_p)$ is the action induced by the standard additive local Clifford group.

For one state, define

$$\begin{aligned} \Gamma_\psi &= \{[U] \in \text{PU}(q^{2m}) : U = U_1 \otimes \cdots \otimes U_{2m}, U|\psi\rangle \in \mathbb{C}|\psi\rangle\}, \\ \text{Aut}_{\text{tr}}(\psi) &= \{(F_i) \in \prod_i \text{Sp}(P_i) : F_j M_A(i, j) = M_A(i, j) F_i \text{ for all } A, i, j \in A\}. \end{aligned}$$

Theorem 1.1 and the Clifford sequence (2.1) give a well-defined homomorphism

$$\lambda_\psi : \Gamma_\psi \longrightarrow \text{Aut}_{\text{tr}}(\psi), \quad [U_1 \otimes \cdots \otimes U_{2m}] \longmapsto (\overline{U}_i)_i,$$

where $\overline{U}_i$ is the induced symplectic label map. A product decomposition is unique up to one-site scalars, which do not change $\overline{U}_i$. The injection $L_\psi \hookrightarrow \Gamma_\psi$ sends a label ℓ to the projective Pauli $[W_\ell]$.

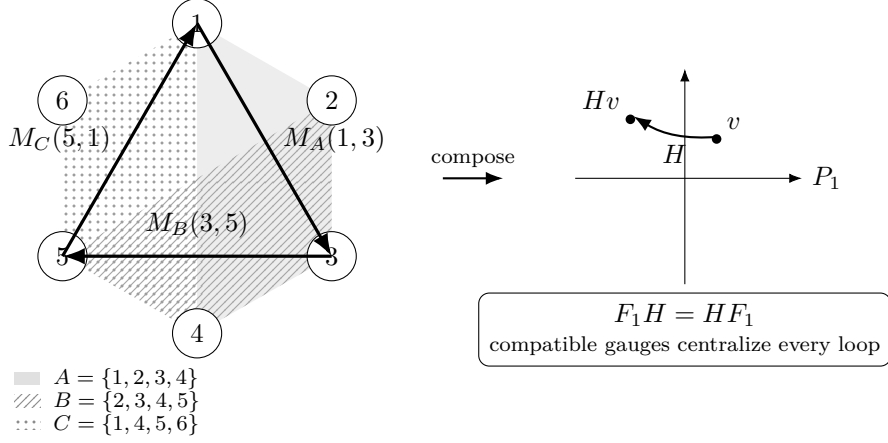


Figure 2: Operator pushing on minimum supports, shown for six parties. Each $(m + 1)$ -party support identifies the local Weyl-label planes of its parties through the bijective projections of the support theorem; overlapping supports compose to a loop whose holonomy $H = M_C M_B M_A$ is a linear map of the base plane, carrying a label v of the base plane to Hv . A local frame change is compatible precisely when it intertwines every transition; for one state over a prime field, its base block centralizes every loop holonomy.

Lemma 4.1 (Pauli phase correction). Let ψ and ϕ be stabilizer states with Lagrangian label groups L_ψ and L_ϕ . If a product Clifford U induces a symplectic label map carrying L_ψ onto L_ϕ , then there is a product Pauli W_v such that $W_v U|\psi\rangle \in \mathbb{C}|\phi\rangle$.

Proof. The state $U|\psi\rangle$ is stabilized on the same Pauli axes as $|\phi\rangle$, possibly with a different stabilizer character. The trace-symplectic pairing identifies the ambient label space modulo $L_\phi = L_\phi^\perp$ with the character dual of L_ϕ . Choose v whose pairing character is the inverse of the discrepancy. Conjugation by the product Pauli W_v cancels it. The corrected state and $|\phi\rangle$ therefore lie in the same one-dimensional joint stabilizer eigenspace. $\square$

Proof of Theorem 1.4. First fix the party labels. A block symplectic map $F = \bigoplus_i F_i$ carrying L_ψ to L_ϕ carries every supported subgroup $L_\psi(A)$ to $L_\phi(A)$. Writing these subgroups as graphs through any projection p_i gives exactly (4.3).

Conversely, suppose (4.3) holds. Fix A and $i \in A$. Every element of $L_\psi(A)$ is uniquely determined by its i -coordinate, and the equations with $j \in A$ show that its image under F is the unique element of $L_\phi(A)$ with i -coordinate $F_i p_i(\ell)$. Thus $FL_\psi(A) = L_\phi(A)$. Since the $(m + 1)$ -supported subgroups generate the full label groups, $FL_\psi = L_\phi$.

Sequence (2.1) supplies local Clifford lifts of the F_i , and Lemma 4.1 supplies the product-Pauli correction. Hence an equivalence of transition systems lifts to an LC equivalence. The reverse implication follows from the induced label action, and Theorem 1.1 replaces LC by LU. A party permutation is absorbed by relabelling first.

For automorphisms, λ_ψ is surjective by the same lifting and phase-correction argument. Its kernel consists of projective product Paulis by (2.1). Such a Pauli fixes the state ray exactly when its label lies in $L_\psi^\perp = L_\psi$. Thus the displayed injection of L_ψ identifies it with $\ker \lambda_\psi$, proving exactness.

Finally suppose $q = p$. Form the directed multigraph with vertices the parties and an edge (A, i, j) carrying $M_A(i, j)$. Every pair of parties belongs to an $(m + 1)$ -set, so the graph is connected. Choose a base b , a spanning tree, and tree transports $T_i : P_b \rightarrow P_i$, with $T_b = I$. For every edge define its based holonomy

$$H_A(i, j) = T_j^{-1} M_A(i, j) T_i \in \text{GL}_2(q).$$

Compatibility propagates a base block F_b uniquely as $F_i = T_i F_b T_i^{-1}$; the edge (A, i, j) is then compatible exactly when F_b commutes with $H_A(i, j)$. Since $\text{Sp}_2(\mathbb{F}_q) = \text{SL}_2(q)$ is normal in $\text{GL}_2(q)$, every propagated block remains symplectic. Evaluation at the base therefore identifies $\text{Aut}_{\text{tr}}(\psi)$ with the holonomy centralizer inside $\text{SL}_2(q)$. Different spanning trees generate the same based holonomy subgroup; changing the base conjugates that subgroup, and hence its centralizer, by a chosen transport. $\square$

4.1 Encoder consequences

Proof of Corollary 1.2. The input marginal of each AME Choi state is $q^{-1}I$, so one-leg reshaping gives an isometry. The projection of its stabilizer label group onto the input Weyl-label space is surjective: otherwise a nonzero local label orthogonal to the image would lie in the maximal isotropic space $L^\perp = L$, contradicting the absence of a one-party stabilizer. The kernel therefore has q^{2m-2} elements and stabilizes the image, making it a one-logical-qudit stabilizer code. AME corrects erasure of any $m-1$ outputs, so its distance is at least m ; the quantum Singleton bound gives equality. This is the standard perfect-tensor/quantum-MDS correspondence used in [Tan26b, Theorems 3.5 and 3.6].

Let $|\Phi\rangle = q^{-1/2} \sum_x |x, x\rangle$, and use the Choi convention $|\psi\rangle = (I \otimes V_\psi)|\Phi\rangle$ and $|\phi\rangle = (I \otimes V_\phi)|\Phi\rangle$. Writing $U_{\text{phys}} = \bigotimes_i U_i$, the relation $U_{\text{phys}} V_\psi = V_\phi L$ gives

$$(I \otimes U_{\text{phys}})|\psi\rangle = (L^T \otimes I)|\phi\rangle.$$

Hence

$$((L^T)^{-1} \otimes U_{\text{phys}})|\psi\rangle = |\phi\rangle.$$

Theorem 1.1 makes every displayed local factor Clifford. Entrywise conjugation preserves the finite-field Weyl group, since $\overline{X(a)} = X(a)$ and $\overline{Z(b)} = Z(-b)$; it therefore preserves the Clifford normalizer. Taking adjoints preserves the normalizer as well, and $U^T = (\overline{U})^\dagger$. Thus transposition preserves the finite-field Clifford group, and L is Clifford. $\square$

5 Cleaning-based quantitative rounding

We now turn from exact equivalence to approximate symmetry. Estimating one reduced operator would lose the exponentially small nonidentity signal of a minimum-support marginal. The global argument avoids that loss. Choose one AME party as a logical input. The other $2m-1$ parties form an $[[2m-1, 1, m]]_q$ stabilizer code, and the product of their local factors approximately implements the transpose-inverse of the factor on the chosen party. Three correctable regions then test that logical action by a nested commutator. Indeed, distance m makes every set of at most $m-1$ erased physical parties correctable, and the $2m-1$ physical parties split into regions of sizes $m-1, m-1, 1$.

The exact mechanism is standard: the qubit stabilizer cleaning lemma is due to Bravyi and Terhal [BT09, Lemma 1]; in the additive prime-power setting used here, the same cleaning step is the elementary symplectic statement recorded in the proof of Lemma 5.1. Nested-commutator localization underlies the Bravyi–König hierarchy theorem [BK13], and Pastawski–Yoshida prove directly that a transversal logical unitary on three cleanable regions is Clifford [PY15, Lemma 5]. Disjointness gives a complementary exact-code hierarchy bound [JOKY18, Theorem 5]. The point below is therefore not a new qualitative transversal-gate obstruction. It is the robust step: an implementation with leakage error ε forces every nested logical Weyl commutator within 8ε of a scalar, after which a Fourier argument gives an explicit Clifford distance.

For an encoding isometry $V : \mathbb{C}^q \rightarrow \mathcal{H}$, write

$$E(T, L) = q^{-1/2} \|TV - VL\|_{\text{HS}}.$$

It satisfies $E(TS, LM) \leq E(T, L) + E(S, M)$ and $E(T^\dagger, L^\dagger) = E(T, L)$.

Lemma 5.1 (Quantitative cleaning commutator). Let V encode one logical qudit in a stabilizer code whose physical parties have a partition into three correctable regions. Let T be transversal, and suppose $E(T, L) \leq \varepsilon$. For all logical Weyl operators P, Q ,

$$q^{-1/2} \min_{|z|=1} \|[[L, P], Q] - zI\|_{\text{HS}} \leq 8\varepsilon, \quad (5.1)$$

where $[A, B] = ABA^\dagger B^\dagger$.

Proof. Let R_1, R_2, R_3 be the assumed correctable partition. The stabilizer cleaning lemma supplies a representative $\tilde{P}$ of P outside R_1 and a representative $\tilde{Q}$ of Q outside R_2 . For reference, cleaning here is the elementary symplectic statement that the restriction of a normalizer label to a correctable region lies in the restriction of the stabilizer label space; subtracting that stabilizer removes the restriction.

Transversality makes $[T, \tilde{P}]$ supported outside R_1 , so $C = [[T, \tilde{P}], \tilde{Q}]$ is supported on $R_1^c \cap R_2^c = R_3$. The composition rules give

$$E(C, [[L, P], Q]) \leq 4\varepsilon. \quad (5.2)$$

Knill–Laflamme correction on R_3 gives $V^\dagger CV = cI$. Hence the logical nested commutator lies within 4ε of cI . Its normalized Hilbert–Schmidt norm is one, so $|c| \geq 1 - 4\varepsilon$. If $c = 0$, then $\varepsilon \geq 1/4$, and (5.1) is immediate because its left-hand side is at most $2 \leq 8\varepsilon$. Otherwise, replacing c by $c/|c|$ proves (5.1). Notice that C need not preserve the code. Only its scalar compression is used, while (5.2) controls leakage. $\square$

The remaining step is finite-dimensional Fourier concentration. It is stated separately because it is the point where scalar commutators become Clifford axes.

Lemma 5.2 (Nested Weyl commutators round to a Clifford). Let $L \in U(q)$, $q = p^e$, and suppose for all Weyl labels v, w that

$$q^{-1/2} \min_{|z|=1} \|[[L, W_v], W_w] - zI\|_{\text{HS}} \leq \eta.$$

If

$$\eta < \tau_p := \min \left\{ \frac{1}{3}, \frac{\sin(\pi/p)}{2\sqrt{2}} \right\},$$

then some additive one-qudit Clifford K satisfies

$$q^{-1/2} \min_{|z|=1} \|L - zK\|_{\text{HS}} \leq \eta.$$

Proof. Put $A_v = LW_vL^\dagger$. Conjugation of W_w by W_v changes only its phase, so every $[A_v, W_w]$ is within η of a scalar. We use the following Fourier observation twice. If $A = \sum_a c_a W_a$ is unitary and every $[A, W_w]$ is within η of a scalar, then A lies within η of one Weyl axis. Indeed, with $p_a = |c_a|^2$, the commutator estimate gives

$$\left| \sum_a p_a \chi_w(a) \right| \geq 1 - \eta^2/2.$$

Character orthogonality, averaged over w , yields $\sum_a p_a^2 \geq (1 - \eta^2/2)^2$. Thus $\max_a p_a \geq (1 - \eta^2/2)^2$, which is precisely the asserted normalized distance bound.

Apply the observation to each A_v , and call its unique nearby Weyl label $F(v)$. The exact identity $A_{v+u} \in \mathbb{C}A_v A_u$ puts the axes labelled by $F(v+u)$ and $F(v)+F(u)$ within 3η . Distinct

Weyl axes have phase-optimized distance $\sqrt{2}$, so $\eta < 1/3$ makes F additive. Comparing the exact commutator phases of A_v, A_u with those of their nearby Weyls costs at most 4η . The least nontrivial p -th-root chord is $2\sin(\pi/p)$, so the stated bound makes F trace-symplectic.

Choose an additive Clifford K_0 inducing F . Then $K_0^\dagger L$ approximately fixes every Weyl axis, so all its Weyl commutators are within η of scalars. The Fourier observation puts it within η of a Weyl; multiplying that Weyl into K_0 gives the required Clifford. $\square$

Two further estimates turn local Clifford frames into the stated global decomposition. The first separates distinct Clifford images of a stabilizer state by a uniform gap.

Lemma 5.3 (Quantized stabilizer overlap). Let $|\psi\rangle, |\phi\rangle$ be stabilizer states on n qudits of local dimension $q = p^e$, with label groups L_ψ, L_ϕ and stabilizer characters χ_ψ, χ_ϕ . Then

$$|\langle\psi|\phi\rangle|^2 = \begin{cases} |L_\psi \cap L_\phi|/q^n, & \text{if } \chi_\psi = \chi_\phi \text{ on } L_\psi \cap L_\phi, \\ 0, & \text{otherwise.} \end{cases}$$

In particular, a product Clifford K either preserves the ray of ψ , or

$$\varepsilon(K) \geq d_p := (2 - 2p^{-1/2})^{1/2}.$$

Proof. Weyl orthogonality and the stabilizer expansions give

$$|\langle\psi|\phi\rangle|^2 = q^{-n} \sum_{\ell \in L_\psi \cap L_\phi} (\chi_\psi \overline{\chi_\phi})(\ell).$$

The summand is a character of the intersection, so the sum is its order when the character is trivial and zero otherwise. Since the intersection has p -power order, every nonunit overlap has modulus at most $p^{-1/2}$. Apply this to $\phi = K\psi$, which is again a stabilizer state. $\square$

The second estimate controls all residual generators at once. Its proof is included because it is a load-bearing part of the headline quantitative theorem; Appendix A gives stronger variants under other hypotheses.

Proposition 5.4 (Residual stability). Let $|\psi\rangle$ be a stabilizer AME($2m, q$) state with $m \geq 2$, and let $V = \bigotimes_j e^{ih_j}$, where every h_j is traceless Hermitian of spectral spread at most $s \leq \pi$. Put

$$c(s) = \max \left\{ 1 - \frac{s^2}{12}, \frac{4}{\pi^2} \right\}.$$

If $D^2 = \sum_j \|h_j\|_F^2 \leq q/4$, then

$$D \leq 2\sqrt{q/c(s)} \varepsilon(V) \leq \pi\sqrt{q} \varepsilon(V).$$

Proof. Expand each factor in the normalized Weyl basis as $e^{ih_j} = \sum_v a_j(v)W_v$, and put $\eta_j = 1 - |a_j(0)|^2$ and $H = \sum_j \eta_j$. Choose an m -party set A . The AME support formula makes restriction of the stabilizer label group L to either A or A^c a bijection. Hence the vectors

$$u(\ell) = \prod_{j \in A} |a_j(\ell_j)|, \quad v(\ell) = \prod_{j \notin A} |a_j(\ell_j)| \quad (\ell \in L)$$

are unit vectors. Expanding the stabilizer expectation and taking moduli termwise gives $\varepsilon(V)^2 \geq \|u - v\|^2$.

For each $j \notin A$ and Weyl label w , let $\ell^{j,w}$ be the unique stabilizer label supported on $A \cup \{j\}$ whose j -coordinate is w . The support formula makes this parametrization bijective and gives

$$\sum_{w \neq 0} v(\ell^{j,w})^2 \geq \eta_j(1 - H), \quad \sum_{w \neq 0} u(\ell^{j,w})^2 \leq \prod_{i \in A} \eta_i.$$

The corresponding families are disjoint as j varies. Applying $(x - y)^2 \geq x^2/2 - y^2$, and then repeating the estimate with A and A^c exchanged, yields

$$\|u - v\|^2 \geq \left(\frac{1}{2}(1 - H) - \eta^{m-1}\right)H, \quad \eta = \max_j \eta_j.$$

For $m \geq 3$ and $H \leq 1/4$, the coefficient is at least $5/16$. For $m = 2$, retaining the four pair products before the maximum estimate gives $H/2 - 3P$, where $P \leq H^2/4$, again at least $5H/16$. Thus in all cases $\varepsilon(V)^2 \geq H/4$.

It remains to compare H with D . If X is uniform on the eigenvalues of h_j , then

$$\eta_j = 1 - |\mathbb{E}e^{iX}|^2 = \mathbb{E}[1 - \cos(X - X')] \geq c(s) \frac{\|h_j\|_F^2}{q},$$

using both $\cos x \leq 1 - x^2/2 + x^4/24$ for $|x| \leq s$ and $1 - \cos x \geq 2x^2/\pi^2$ for $|x| \leq \pi$. Conversely $\eta_j \leq \|h_j\|_F^2/q$, so $H \leq D^2/q \leq 1/4$. Combining the inequalities gives $\varepsilon(V)^2 \geq c(s)D^2/(4q)$, which proves the claim. $\square$

At $\varepsilon = 0$, the first two lemmas in this section recover the three-cleanable-region Clifford conclusion of [PY15, Lemma 5]. Their quantitative combination, and the constants in (5.1), are what will be used here.

Theorem 5.5 (Cleaning-based global rounding). Let $|\psi\rangle$ be a stabilizer AME($2m, q$) state, $q = p^e$, $m \geq 2$, and $n = 2m$. Put

$$d_p = (2 - 2p^{-1/2})^{1/2},$$

and

$$R_{n,q}^{\text{clean}} = \min \left\{ \frac{\tau_p}{8}, \frac{1}{4\sqrt{2q}}, \frac{1}{8\pi\sqrt{n}}, \frac{d_p}{1 + 4\pi\sqrt{n}} \right\}. \quad (5.3)$$

If $\varepsilon(U) < R_{n,q}^{\text{clean}}$, then, up to a global phase,

$$U = g \bigotimes_{i=1}^n e^{ih_i}, \quad g \in G(\psi),$$

where the h_i are traceless Hermitian, have spectral spread at most π , and satisfy

$$\left(\sum_i \|h_i\|_F^2 \right)^{1/2} \leq \pi\sqrt{q}\varepsilon(U). \quad (5.4)$$

Before the exact-symmetry step, the recovered local Cliffords obey

$$q^{-1/2} \min_{|z|=1} \|U_i - zK_i\|_{\text{HS}} \leq 8\varepsilon(U) \quad (5.5)$$

at every party. For fixed q , (5.3) has order $n^{-1/2}$, conditional on existence of the stated AME states.

Proof. Local Clifford frames. Choose party i as the logical leg and write the AME Choi form as $q^{-1/2} \sum_a |a\rangle \otimes V_i|a\rangle$. The other local factors form a transversal physical unitary T_i , and vectorization gives a logical unitary L_i , obtained from U_i by transpose and inverse, with $E(T_i, L_i) = \varepsilon(U)$. Lemmas 5.1 and 5.2, with $\eta = 8\varepsilon(U)$, give (5.5); transpose and inverse preserve the additive Clifford group.

Centered residual generators. Put $K = \bigotimes_i K_i$. For each i , choose a phase z_i attaining the minimum in (5.5), and let a_i be the principal Hermitian logarithm defined by

$$z_i^{-1} K_i^\dagger U_i = e^{ia_i}.$$

Set $\bar{a}_i = q^{-1} \text{Tr}(a_i)$ and $h_i = a_i - \bar{a}_i I$. Rephase K_i by $z_i e^{i\bar{a}_i}$; this changes $K = \bigotimes_i K_i$ only by a global phase and does not change its projective Clifford action. After this rephasing, $K_i^\dagger U_i = e^{ih_i}$, each h_i is traceless, centering does not increase its Frobenius norm, and its spectral spread is unchanged. Let $D^2 = \sum_i \|h_i\|_F^2$. The chord-angle bound gives

$$D^2 \leq \frac{\pi^2 q}{4} \sum_i \left(q^{-1/2} \min_{|z|=1} \|U_i - z K_i\|_{\text{HS}} \right)^2 \leq 16\pi^2 q n \varepsilon(U)^2. \quad (5.6)$$

The second clause of (5.3) puts every spectrum in an arc of length π ; the third gives $D \leq \sqrt{q}/2$.

Exact branch selection. Two-uniformity gives $\|(K^\dagger U - I)\psi\| \leq D/\sqrt{q} \leq 4\pi\sqrt{n}\varepsilon(U)$. The fourth clause and the triangle inequality put $K\psi$ less than d_p from ψ , so Lemma 5.3 makes K an exact symmetry. Apply Proposition 5.4 to obtain (5.4). $\square$

The theorem already contains a genuinely collective estimate once the exact branch has been selected. Writing $g = \bigotimes_i g_i$ in the factorization above,

$$\sum_i \left(q^{-1/2} \min_{|z|=1} \|U_i - z g_i\|_{\text{HS}} \right)^2 \leq \frac{D^2}{q} \leq \pi^2 \varepsilon(U)^2. \quad (5.7)$$

The first inequality is the elementary chord bound $\|e^{ih_i} - I\|_{\text{HS}} \leq \|h_i\|_F$. Thus the $\sqrt{n}$ loss is absent after exact-symmetry recovery.

The same estimate applies to intertwiners between two states as soon as one exact intertwiner is fixed. This separates the cost of finding the correct orbit from stability within that orbit.

Corollary 5.6 (Relative intertwiner rounding). Let $|\psi\rangle, |\phi\rangle$ be stabilizer $\text{AME}(2m, q)$ states as in Theorem 5.5, and suppose that $B = \bigotimes_i B_i$ is an exact product-unitary intertwiner from ψ to ϕ . For $U = \bigotimes_i U_i$, put

$$\varepsilon_{\psi \rightarrow \phi}(U) = \min_{|z|=1} \|U\psi - z\phi\|.$$

If $\varepsilon_{\psi \rightarrow \phi}(U) < \tau_p/8$, then each U_i is within normalized Hilbert–Schmidt distance $8\varepsilon_{\psi \rightarrow \phi}(U)$ of $B_i K_i$ for a one-qudit Clifford K_i . If, in addition, $\varepsilon_{\psi \rightarrow \phi}(U) < R_{n,q}^{\text{clean}}$, then, up to phase,

$$U = H \bigotimes_i e^{ih_i}, \quad H\psi \in \mathbb{C}\phi, \quad \left(\sum_i \|h_i\|_F^2 \right)^{1/2} \leq \pi\sqrt{q}\varepsilon_{\psi \rightarrow \phi}(U),$$

where H is an exact product-unitary intertwiner and the h_i are traceless Hermitian of spectral spread at most π . After branch selection,

$$\sum_i \left(q^{-1/2} \min_{|z|=1} \|U_i - z H_i\|_{\text{HS}} \right)^2 \leq \pi^2 \varepsilon_{\psi \rightarrow \phi}(U)^2.$$

Thus the stability constants are independent of the party count; the certified radius remains $R_{n,q}^{\text{clean}}$.

Proof. Set $W = B^\dagger U = \bigotimes_i B_i^\dagger U_i$. Since $B\psi$ and ϕ span the same ray, $\varepsilon(W) = \varepsilon_{\psi \rightarrow \phi}(U)$. The first paragraph of the proof of Theorem 5.5 gives the local 8ε estimate under $\varepsilon(W) < \tau_p/8$. Under the stronger radius hypothesis, apply the theorem and (5.7) to W . If $W = g \bigotimes_i e^{ih_i}$ with $g \in G(\psi)$, then $H = Bg$ is an exact intertwiner from ψ to ϕ , and left multiplication by each B_i preserves normalized Hilbert–Schmidt distance. This proves all three estimates. Exact rigidity also makes every B_i Clifford, but that fact is not needed for the reduction. $\square$

This does not enlarge the entry radius in (5.3). Before the branch is known, cleaning supplies only the separate bounds (5.5) for locally chosen Cliffords K_i . The n encoded implementation errors are unitary reshaping of the same global defect vector, not projections onto orthogonal sectors, so their squared bounds cannot be summed by a Bessel argument. A radius-improving collective estimate would itself imply constant-radius branch selection: the chord–angle inequality would make $\sum_i \|h_i\|_F^2 = O(q\varepsilon^2)$, and the uniform stabilizer gap would force $\bigotimes_i K_i$ to be exact. The missing input is therefore compatibility of the recovered discrete Clifford data, not a sharper final norm summation.

The linear part of that compatibility can in fact be recovered without a party-count loss.

Proposition 5.7 (Robust compatibility of the transition maps). Let $|\psi\rangle$ be a stabilizer $\text{AME}(2m, q)$ state, $q = p^e$, and let $U = \bigotimes_i U_i$ be a product unitary. Suppose

$$\varepsilon(U) < R_q^{\text{trans}} := \min \left\{ \frac{\tau_p}{8}, \frac{2 \sin(\pi/p)}{4 + 64\sqrt{q}} \right\}.$$

The cleaning and Fourier lemmas give local Cliffords K_i satisfying (5.5). Let F_i be their symplectic maps. Then

$$F_j M_A(i, j) = M_A(i, j) F_i$$

for every $(m+1)$ -set A and $i, j \in A$. Consequently $F = \bigoplus_i F_i$ preserves the stabilizer label group L , and some product-Pauli correction $W_v K$ is an exact symmetry of ψ .

Proof. Fix A , distinct $i, j \in A$, and put $B = A^c \cup \{i, j\}$. Both A and B have size $m+1$, while $A \cap B = \{i, j\}$. Choose $\ell \in L(A)$ and $r \in L(B)$. Using the stabilizer lifts $\widetilde{W}_\ell$ and $\widetilde{W}_r$ from (2.2), set

$$C = [U \widetilde{W}_\ell U^\dagger, \widetilde{W}_r] = [U W_\ell U^\dagger, W_r],$$

where the second equality holds because scalar lift phases cancel from a commutator. After choosing the phase of U realizing its defect, $\|(U \widetilde{W}_\ell U^\dagger - I)\psi\| \leq 2\varepsilon(U)$, and the same holds with the adjoint. Since $\widetilde{W}_r \psi = \psi$,

$$\|(C - I)\psi\| \leq 4\varepsilon(U). \quad (5.8)$$

The commutator is supported only on $A \cap B = \{i, j\}$. Choose phases so that $\|U_s - K_s\|_{\text{HS}} \leq 8\sqrt{q}\varepsilon(U)$. Replacing $U_s W_{\ell_s} U_s^\dagger$ by its K_s -conjugate costs at most $16\sqrt{q}\varepsilon(U)$ in operator norm, and replacing its commutator with W_{r_s} costs at most twice that. On the two sites, the total cost is therefore at most $64\sqrt{q}\varepsilon(U)$. The Clifford commutator is a scalar p -th root ζ , so (5.8) gives

$$|\zeta - 1| \leq (4 + 64\sqrt{q})\varepsilon(U) < 2 \sin(\pi/p).$$

Hence $\zeta = 1$.

Write $a = \ell_i$, $b = r_i$, $\ell_j = M_A(i, j)a$, and $r_j = M_B(i, j)b$. The identity $\zeta = 1$, for all a, b , says

$$[F_i a, b] + [F_j M_A(i, j)a, M_B(i, j)b] = 0. \quad (5.9)$$

Here brackets denote the trace-symplectic pairing. Isotropy of L , applied to the element of $L(A)$ whose i -coordinate is $F_i a$, gives

$$[F_i a, b] + [M_A(i, j) F_i a, M_B(i, j) b] = 0.$$

Subtracting this from (5.9), using surjectivity of $M_B(i, j)$ and nondegeneracy of the pairing, proves the transition equation. The exact transition-map theorem gives $FL = L$, and Lemma 4.1 supplies W_v . $\square$

The uncontrolled Pauli correction does not obstruct the logical action of a chosen encoder.

Corollary 5.8 (Logical Clifford rounding). Under the hypotheses and radius of Proposition 5.7, choose party i as the logical leg, and let L_i be the logical unitary induced by the other $2m-1$ factors as in the proof of Theorem 5.5. There is an exactly transversally realizable logical Clifford L'_i such that

$$q^{-1/2} \min_{|z|=1} \|L_i - z L'_i\|_{\text{HS}} \leq 8\varepsilon(U).$$

Proof. Write the exact symmetry supplied by the proposition as $W_v K$, where $K = \bigotimes_j K_j$. The projection of the stabilizer label group onto the Weyl plane at party i is surjective: this already holds for every minimum-support subgroup containing i . Multiply $W_v K$ by a stabilizer whose i -th label is $-v_i$. The result is still an exact product-Clifford symmetry, and its factor at i is projectively K_i . The Choi correspondence therefore makes its other factors an exact transversal implementation of $L'_i = (K_i^\top)^{-1}$. The logical unitary L_i is the same transpose-inverse transform of U_i , up to phase, so (5.5) proves the displayed bound. $\square$

The Pauli correction in the proposition is essential. All localized commutators are unchanged when a recovered K_i is multiplied by a local Weyl: that multiplication changes conjugation phases, and the commutator cancels them. Equivalently, the transition maps see L but not its stabilizer character. The correction v can therefore be nonzero modulo L , in which case $W_v K$ need not be locally close to U . Exact affine compatibility cannot be obtained from these commutator tests, and Proposition 5.7 alone does not improve (5.3). Controlling the character discrepancy is the same remaining branch-selection problem isolated by the collective estimate above.

The exact formula, rather than a fixed-dimension slogan, controls scaling. Uniformly over prime powers $q = p^e$ and existing tensors,

$$R_{n,q}^{\text{clean}} = \Theta\left(\min\{p^{-1}, q^{-1/2}, n^{-1/2}\}\right), \quad (5.10)$$

with absolute implied constants. Indeed $\sin(\pi/p) = \Theta(p^{-1})$, while d_p is bounded above and below by positive absolute constants. An asymptotic use of (5.10) must specify an existing family.

For the explicit generalized and extended generalized Reed–Solomon construction [SR86], together with the standard MDS-to-AME construction [RGRA18], $n = 2m \leq q + 1$, so the party-count term cannot be smaller than the local-dimension term. Hence

$$R_{n,q}^{\text{clean}} = \begin{cases} \Theta(q^{-1}), & q = p, \\ \Theta(q^{-1/2}), & q = p^e, e \geq 2. \end{cases} \quad (5.11)$$

On a length-saturating subfamily $n \asymp q$, these are $\Theta(n^{-1})$ and $\Theta(n^{-1/2})$, respectively. For fixed characteristic and growing extension degree, the second line applies. The earlier fixed- q , $n^{-1/2}$ reading of the exact minimum remains a conditional formal slice only; it is not an AME-existence claim.

If the sharper local constant and the old global generator budget are needed, the same proof gives the conclusion of Corollary A.19 under

$$\varepsilon(U) < \min \left\{ \frac{\tau_p}{8}, \frac{1}{16\pi n\sqrt{q}} \right\}. \quad (5.12)$$

Indeed (5.5), the operator-norm bound by Hilbert–Schmidt norm, and the principal logarithm give $\sum_i \|h_i\|_{\text{op}} \leq 8\pi n\sqrt{q}\varepsilon(U) < 1/2$; the sharp local theorem then gives $D \leq \sqrt{6q/5}\varepsilon(U)$.

The cleaning route supersedes both explicit radii above on every admissible parameter pair where its displayed minimum is larger, and it has no boundary at $q = 4$. The earlier theorems remain useful because they expose the exact cost of reading one marginal and of aggregating orthogonal minimum-support sectors. The new $n^{-1/2}$ scale is a proved lower bound, not an optimality claim; no product-unitary family is known to saturate the collective frame-error estimate (5.6).

6 Conclusion

The same support geometry controls the exact and approximate problems. On an $(m+1)$ -party marginal it exposes every local Weyl axis, which turns arbitrary product-unitary intertwiners into product Cliffords. The transition maps between the resulting local Weyl frames organize the residual freedom. After one party is reinterpreted as a logical input, the support geometry becomes three-region correctability. Cleaning and Weyl–Fourier concentration then recover the same discrete Clifford frame without paying the exponentially small amplitude of a single marginal.

The quantitative conclusion is uniform only when all varying parameters are displayed:

$$R_{n,q}^{\text{clean}} = \Theta\left(\min\{p^{-1}, q^{-1/2}, n^{-1/2}\}\right), \quad q = p^e.$$

For fixed local dimension this gives the conditional $n^{-1/2}$ scale along any existing stabilizer-AME family. On generalized and extended Reed–Solomon constructions, where $n \leq q + 1$, the scale is $\Theta(q^{-1})$ over prime fields and $\Theta(q^{-1/2})$ when $e \geq 2$. These are certified radii; their optimal order is open.

The robust transition-map result identifies the remaining bottleneck. Linear symplectic compatibility becomes exact at a radius depending only on the local dimension, but commutators discard the affine Weyl phases. They therefore determine the stabilizer label group and not its character. The logical action of any chosen encoder can nevertheless be rounded within 8ε to an exactly transversally realizable Clifford, because a stabilizer cancels the Pauli correction on the logical leg. A party-count-independent global entry theorem would need a new way to control that character discrepancy, or an equivalent collective branch selection principle, before the product-Pauli correction is known to be locally small.

A Secondary quantitative routes and local stability

This appendix compares mechanisms that are not needed to prove the main cleaning argument. It first derives discreteness and local quadratic stability from two-uniformity alone. It then shows how higher uniformity enlarges the certified generator-coordinate region, strengthens the residual estimate proved directly in Section 5, and records weaker single-marginal and aggregate rounding routes. The final subsection translates the results to local gauge transformations of two-unitary gates. Readers interested only in the headline exact and quantitative theorems may skip the comparisons.

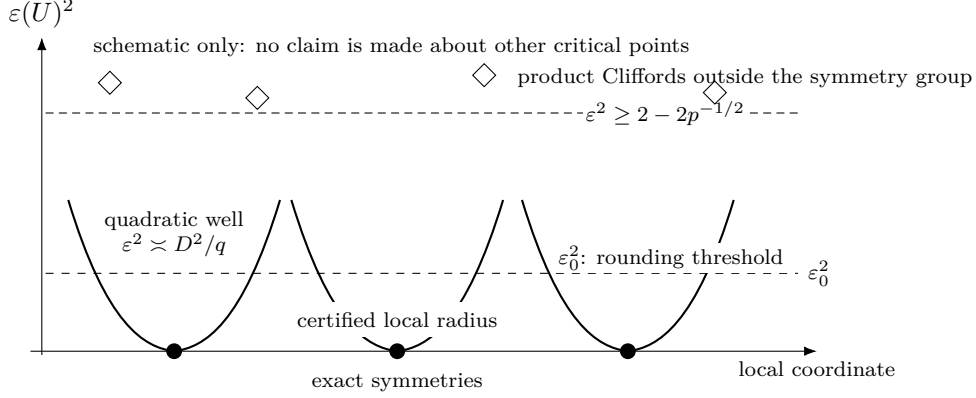


Figure 3: Roadmap for the defect estimates (a one-dimensional schematic of a high-dimensional product-unitary group). Exact symmetries are isolated zeros with certified local quadratic growth. Every product Clifford outside the symmetry group has defect at least $(2 - 2p^{-1/2})^{1/2}$; below the explicit decomposition threshold, an arbitrary product unitary rounds to a Clifford and then to an exact symmetry. The picture asserts neither that every local minimum is Clifford nor that the displayed coordinate exhausts the group.

A.1 Discreteness from two-party maximal mixing

The rigidity theorem uses the stabilizer hypothesis to identify the symmetry group. Finiteness of that group needs far less: two-party marginals alone, at any local dimension, with no stabilizer structure. Throughout this subsection $q \geq 2$ is an arbitrary integer; only the statements naming Clifford operators return to $q = p^e$.

Call a pure state $|\psi\rangle \in (\mathbb{C}^q)^{\otimes n}$ *2-uniform* if

$$\rho_{\{j,k\}}^\psi = q^{-2}I \quad \text{for all } j \neq k.$$

Tracing once more makes every one-party marginal $q^{-1}I$. Such states exist only for $n \geq 4$, since at $n = 3$ purity forces $\text{rank } \rho_{\{1,2\}} = \text{rank } \rho_{\{3\}} \leq q$; and every $\text{AME}(2m, q)$ state with $m \geq 2$ is 2-uniform, stabilizer or not. Write

$$G(\psi) = \{U_1 \otimes \cdots \otimes U_n : (U_1 \otimes \cdots \otimes U_n)|\psi\rangle \in \mathbb{C}|\psi\rangle\}$$

for its product-unitary symmetry group and, for a product unitary U ,

$$\varepsilon(U) = \min_{\theta} \|U|\psi\rangle - e^{i\theta}|\psi\rangle\| = (2 - 2|\langle\psi|U|\psi\rangle|)^{1/2}$$

for its defect. Because operators on distinct sites commute, a product unitary is a single exponential: $\bigotimes_j e^{iH_j} = e^{iL}$ with $L = \sum_j H_j^{(j)}$, where $H^{(j)}$ denotes H acting at site j . Splitting each H_j into a traceless part h_j and a multiple of the identity leaves $L = M + cI$ with $M = \sum_j h_j^{(j)}$. Everything below rests on one computation about M .

Figure 3 gives a roadmap for the local estimates, the Clifford separation step, and the resulting explicit threshold.

Lemma A.1 (Local-generator isometry). Let $|\psi\rangle$ be 2-uniform and let h_j and h'_j , $1 \leq j \leq n$, be traceless Hermitian operators on $\mathbb{C}^q$. Put $M = \sum_j h_j^{(j)}$ and $M' = \sum_j h'_j{}^{(j)}$. Then $\langle\psi|M|\psi\rangle = 0$ and

$$\langle M|\psi\rangle, M'|\psi\rangle\rangle = \frac{1}{q} \sum_{j=1}^n \text{Tr}(h_j h'_j). \quad (\text{B.1})$$

In particular $\|M|\psi\rangle\|^2 = q^{-1} \sum_j \|h_j\|_F^2$, so $(h_1, \dots, h_n) \mapsto \sqrt{q} M|\psi\rangle$ is an isometry from the traceless local generators, carrying the product Frobenius form, into the Hilbert space regarded as a real inner product space.

Proof. The one-party marginals give $\langle M \rangle = \sum_j \text{Tr}(\rho_{\{j\}}^\psi h_j) = \sum_j \text{Tr}(h_j)/q = 0$. Expand $\langle \psi | MM' | \psi \rangle$. The terms with equal site indices contribute $\text{Tr}(\rho_{\{j\}}^\psi h_j h'_j) = \text{Tr}(h_j h'_j)/q$. A term with $j \neq k$ contributes $\text{Tr}(\rho_{\{j,k\}}^\psi (h_j \otimes h'_k)) = \text{Tr}(h_j) \text{Tr}(h'_k)/q^2 = 0$; both maximal mixedness of the pair marginal and tracelessness are used. Hence $\langle \psi | MM' | \psi \rangle = q^{-1} \sum_j \text{Tr}(h_j h'_j)$, which is real. Since M and M' are Hermitian, the imaginary part of $\langle M\psi, M'\psi \rangle$ is $\langle [M, M'] \rangle / 2i$, and it vanishes as well: operators at distinct sites commute, and the surviving one-site commutators are traceless, so each term is of the form $\text{Tr}(\rho_{\{j\}}^\psi [h_j, h'_j]) = \text{Tr}([h_j, h'_j])/q = 0$. A real symmetric bilinear form agreeing with the product Frobenius form is that form. $\square$

The identity has no error term, and that is the content of the 2-uniformity hypothesis rather than a coincidence: the cross terms cancel exactly because the pair marginals are exactly $q^{-2}I$. A Bell pair is the boundary case. There $\rho_{\{1,2\}}$ is a pure state, the cross term survives, and $M = h \otimes I - I \otimes h^T$ annihilates the state for every h — the $U \otimes \bar{U}$ gauge line that makes $m \geq 2$ sharp in Theorem 1.1.

The second ingredient is that a one-parameter group of product unitaries has a generator of summed local form. This is not automatic: a one-parameter subgroup of $G(\psi)$ is a priori only a continuous homomorphism into $U(q^n)$, and the factors of a product unitary are determined only up to phases, so differentiating factor by factor carries a lifting obligation.

Lemma A.2 (Product generators). Let $\pi : U(q)^n \rightarrow U(q^n)$ send $(U_1, \dots, U_n)$ to $U_1 \otimes \dots \otimes U_n$. Then π is a smooth homomorphism of compact Lie groups, its image P is a closed subgroup of $U(q^n)$, and

$$\text{Lie}(P) = \left\{ i \sum_j H_j^{(j)} : H_j = H_j^\dagger \text{ on } \mathbb{C}^q \right\}. \quad (\text{B.2})$$

Hence every one-parameter subgroup of a closed subgroup of P has a generator of that form.

Proof. The tensor product of unitaries is multiplicative factor by factor, so π is a smooth homomorphism. Its source is compact, so P is compact and therefore closed, and Cartan's theorem makes P an embedded Lie subgroup of $U(q^n)$. A surjective homomorphism of Lie groups has surjective differential, so $\text{Lie}(P) = d\pi(\mathfrak{u}(q)^n)$, which is the displayed space; its kernel is $\{(i\tau_1 I, \dots, i\tau_n I) : \sum_j \tau_j = 0\}$. A closed subgroup $H \leq P$ is a Lie subgroup with $\text{Lie}(H) \subseteq \text{Lie}(P)$, and the generator of a one-parameter subgroup of H lies in $\text{Lie}(H)$. $\square$

Theorem A.3 (Discreteness from 2-uniformity). Let $|\psi\rangle$ be a 2-uniform pure state of n qudits of local dimension q . Then $\text{Lie}(G(\psi)) = i\mathbb{R}I$, and $G(\psi)/U(1) \cdot I$ is finite. In particular every $\text{AME}(2m, q)$ state with $m \geq 2$, stabilizer or not, has finite product-unitary symmetry group modulo global phase.

Proof. The set $\{U \in U(q^n) : U|\psi\rangle \in \mathbb{C}|\psi\rangle\}$ is a closed subgroup, so $G(\psi) = P \cap \{\dots\}$ is a closed subgroup of P and hence a compact Lie group. Let $X \in \text{Lie}(G(\psi))$. By Lemma A.2, $X = iL$ with $L = \sum_j H_j^{(j)}$ Hermitian; write $L = M + cI$ as above. Each e^{itL} fixes the ray of $|\psi\rangle$, so $e^{itL}|\psi\rangle = \lambda(t)|\psi\rangle$ with $\lambda(t) = \langle \psi | e^{itL} | \psi \rangle$ unimodular and smooth in t . Differentiating at $t = 0$ shows that $|\psi\rangle$ is an eigenvector of the Hermitian operator L , say $L|\psi\rangle = \mu|\psi\rangle$ with μ real. Pairing with $|\psi\rangle$ and using $\langle M \rangle = 0$ gives $\mu = c$, hence $M|\psi\rangle = 0$. By Lemma A.1, $\sum_j \|h_j\|_F^2 = 0$, so every $h_j = 0$ and every H_j is scalar. Thus $\text{Lie}(G(\psi)) = i\mathbb{R}I$. Since $U(1) \cdot I$ is a closed central one-dimensional subgroup of $G(\psi)$, the compact Lie group $G(\psi)/U(1) \cdot I$ has dimension zero and is therefore finite. $\square$

Remark A.4 (Which torus is divided out). Upstairs in $U(q)^n$ the preimage $\pi^{-1}(G(\psi))$ always contains the local-phase torus $T = U(1)^n$, of dimension n . This does not contradict the theorem: $\pi(T) = U(1) \cdot I$ and $\ker \pi \subset T$ has dimension $n-1$, so $\pi^{-1}(G(\psi))/T \cong G(\psi)/U(1) \cdot I$. Statements about local factors below are statements on $\text{PU}(q)^n$, not statements modulo one global phase.

Remark A.5 (Discreteness in the literature). Finiteness of this kind is known in two special cases. Wirthmüller determines the identity component of the local-unitary automorphism group of a binary stabilizer code and gives a protected-qubit criterion for finiteness of the induced group [Wir11, Theorem 7 and Corollary 11]; his exclusion of Bell factors is the $n \geq 4$ boundary above. Englbrecht and Kraus determine the local symmetries of qubit stabilizer states, including where continuous ones occur [EK20]. Tan computes the local symmetry group of the four-qutrit AME state outright, of order 5832 [Tan26b, Theorem 5.3]. Theorem A.3 covers arbitrary local dimension and arbitrary, in particular nonstabilizer, 2-uniform states; to our knowledge no statement at that scope is available.

The hypothesis is exactly 2-uniformity, and one-party marginals are not enough. Słowiak, Sawicki, and Maciążek construct locally maximally entangled states — every one-party marginal maximally mixed — whose local-unitary symmetry group contains a nontrivial continuous image of $\text{SU}(N)$ and is therefore positive-dimensional, with dimension growing in the number of parties [SSM21, Lemma 2]. Passing from one-party to two-party marginals is what forces discreteness. In the language of geometric invariant theory the theorem says that a 2-uniform state is a stable point for $\text{SL}(q, \mathbb{C})^n$ rather than merely a critical one; stabilizer dimensions there are known for generic states and in the extremal qubit case [BRVR18, BLRVR19, LWB08, GKW17], but a genericity statement does not decide a particular state.

Remark A.6 (Two-uniformity does not identify the group). An explicit state shows that the second half of the rigidity phenomenon needs strictly more than the first. Let $q = 2$ and $C = \text{RM}(1, 4) \leq \mathbb{F}_2^{16}$, whose nonzero codeword weights are 8 and 16 and whose dual $\text{RM}(2, 4)$ has minimum distance 4. The equal-phase CSS state of C has label group $C_X \oplus C_Z^\perp$, so by (2.3) a marginal fails to be maximally mixed only when a nonzero word of C or of $C^\perp$ is supported inside it; both distances exceed 2, so the state is 2-uniform and Theorem A.3 makes its symmetry group finite. That group is not Clifford. Writing $T = \text{diag}(1, e^{i\pi/4})$, the product $T^{\otimes 16}$ multiplies $|x\rangle$ by $e^{i\pi \text{wt}(x)/4}$, which is 1 at every codeword weight, so $T^{\otimes 16}$ fixes the state exactly; and $TX(1)T^\dagger = 2^{-1/2}(X(1) + iX(1)Z(1))$ is not a Weyl axis. Two-uniformity buys finiteness and nothing else. Identifying the finite group as local Clifford is what the AME hypothesis and Theorem 1.1 supply, and the next corollary is that identification.

Corollary A.7 (Discrete product-unitary symmetry). Fix a stabilizer AME state $|\psi\rangle$ as in Theorem 1.1, and define the local-factor tuple groups

$$G = \{(U_i) \in U(q)^{2m} : (\bigotimes_i U_i)|\psi\rangle \in \mathbb{C}|\psi\rangle\},$$

$$\tilde{G} = \{((U_i), \pi) \in U(q)^{2m} \rtimes S_{2m} : (\bigotimes_i U_i)\pi|\psi\rangle \in \mathbb{C}|\psi\rangle\},$$

where S_{2m} permutes the local factors. Let $T = (S^1)^{2m}$ be the coordinatewise scalar subgroup and put $\Gamma = G/T$, $\tilde{\Gamma} = \tilde{G}/T$. Then

$$1 \longrightarrow T \longrightarrow G \longrightarrow \Gamma \longrightarrow 1, \quad 1 \longrightarrow T \longrightarrow \tilde{G} \longrightarrow \tilde{\Gamma} \longrightarrow 1$$

are short exact sequences with closed connected kernel and finite discrete quotient; both inclusions of T are continuous. In particular, G and $\tilde{G}$ are finite unions of connected components, each a translate of T . Thus the local-factor automorphism group modulo one-site scalar phases

is finite and discrete, and its identity component before quotienting is exactly T . If Π is the subgroup of party permutations realized by $\tilde{G}$, there is a further exact sequence

$$1 \longrightarrow \Gamma \longrightarrow \tilde{\Gamma} \longrightarrow \Pi \longrightarrow 1.$$

This last extension splits exactly when the realized permutations admit a homomorphic choice of projective product-unitary lifts. We use only the finiteness statement here; the extension class is not part of the quantitative argument.

Proof. Every local factor in a tuple from G or $\tilde{G}$ is Clifford. The projective single-qudit Clifford group over a finite field is finite, as is the party-permutation group. The kernel of projectivizing each local factor is exactly the torus of one-site scalar phases. That torus is the identity component, hence is closed; the finite Hausdorff quotient is discrete. Translation identifies every connected component with the scalar torus, so finitely many quotient representatives cover the group. Forgetting the local factors maps $\tilde{\Gamma}$ onto the realized permutation subgroup Π . Its kernel consists exactly of the fixed-party projective automorphisms, which gives the third sequence. A splitting is, by definition, a group-homomorphic right inverse; arbitrary or generator-by-generator lifts do not provide one. $\square$

A.2 Local quadratic stability

Discreteness makes the exact symmetries isolated. The same identity (B.1) says how fast the defect grows away from them, with a constant that does not involve the number of parties.

Theorem A.8 (Local stability). Let $|\psi\rangle$ be 2-uniform, let each h_j be traceless Hermitian on $\mathbb{C}^q$, and put

$$U = \bigotimes_{j=1}^n e^{ih_j}, \quad t = \sum_{j=1}^n \|h_j\|_{\text{op}}, \quad D = \left(\sum_{j=1}^n \|h_j\|_F^2 \right)^{1/2}.$$

If $t \leq \frac{1}{2}$ then

$$\frac{D^2}{q} \left(1 - \frac{t}{3} \right) \leq \varepsilon(U)^2 \leq \frac{D^2}{q} \left(1 + \frac{t}{3} \right). \quad (\text{B.3})$$

In particular $D \leq \sqrt{6q/5} \varepsilon(U) < 1.096\sqrt{q} \varepsilon(U)$, and $\varepsilon(U)^2/D^2 \rightarrow q^{-1}$ as $t \rightarrow 0$, uniformly in n .

Proof. Since the sites commute, $U = e^{iM}$ with $M = \sum_j h_j^{(j)}$, and $\|M\|_{\text{op}} \leq t$ by the triangle inequality. Write $\langle \cdot \rangle = \langle \psi | \cdot | \psi \rangle$. Taylor's formula with integral remainder gives $e^{ix} = 1 + ix - x^2/2 + r(x)$ with $|r(x)| \leq |x|^3/6$ for real x . Let $M = \sum_i \lambda_i P_i$ be the spectral decomposition and put $p_i = \langle P_i \rangle \geq 0$, so that $\sum_i p_i = 1$. Then $\langle r(M) \rangle = \sum_i p_i r(\lambda_i)$ and

$$|\langle r(M) \rangle| \leq \frac{1}{6} \sum_i p_i |\lambda_i|^3 \leq \frac{t \langle M^2 \rangle}{6}.$$

By Lemma A.1, $\langle M \rangle = 0$ and $\langle M^2 \rangle = D^2/q$, so $\langle e^{iM} \rangle = 1 - \langle M^2 \rangle/2 + \langle r(M) \rangle$. Also $\langle M^2 \rangle \leq t^2 \leq \frac{1}{4}$, so $1 - \langle M^2 \rangle/2 > 0$ and

$$\left| |\langle e^{iM} \rangle| - \left(1 - \frac{1}{2} \langle M^2 \rangle \right) \right| \leq \frac{t \langle M^2 \rangle}{6},$$

using $|\langle e^{iM} \rangle| \geq \text{Re} \langle e^{iM} \rangle$ for the lower estimate. Since $\varepsilon(U)^2 = 2 - 2|\langle e^{iM} \rangle|$, this is exactly (B.3). At $t \leq \frac{1}{2}$ the left inequality reads $\varepsilon^2 \geq \frac{5}{6} D^2/q$, and the two-sided form gives the limit. $\square$

The constant $\sqrt{6/5} = 1.0954\dots$ is therefore sharp to within about ten percent: the true asymptotic constant is $\sqrt{q}$, and the excess is the cubic Taylor cutoff, not structure.

Remark A.9 (The stability constant is a Fisher metric). Identity (B.1) also explains why the constant does not involve n . For the family $U(s) = \bigotimes_j e^{ish_j} = e^{isM}$, the quantum Fisher information of a pure state is $4 \operatorname{Var}_\psi(M)$ [BC94], so

$$F_Q = \frac{4}{q} \sum_{j=1}^n \|h_j\|_F^2.$$

Splitting the variance of a sum of one-site terms into single-party variances and cross-party covariances is the standard framework for metrologically useful entanglement [HLK⁺12, Tó12]; 2-uniformity is exactly the hypothesis that annihilates every covariance and pins every single-party variance. On traceless local generators — and only there, since (B.1) says nothing about generators that are not sums of one-site terms — the Fisher form is a flat multiple of the Euclidean one. A degenerate direction would be a continuous product symmetry, which Theorem A.3 excludes; an enhanced direction, of the kind a GHZ state supplies, would need the pair correlations that 2-uniformity forbids. The stability constant is the inverse of that flat metric.

The Fisher form is quadratic data, so it governs the constant and says nothing about how large a neighbourhood the quadratic estimate covers. That size is fixed by the higher moments of M , and Theorem A.12 and Proposition A.15 below determine it.

A.3 Higher uniformity and the generator-coordinate region

The size of the certified region is controlled by the state's uniformity order, not by the number of parties. Call $|\psi\rangle$ *k-uniform* if every reduced operator on at most k parties is maximally mixed; 2-uniformity is the case $k = 2$, and an $\operatorname{AME}(2m, q)$ state is m -uniform. The hypothesis $t \leq \frac{1}{2}$ in Theorem A.8 pays for a cubic Taylor remainder. Each additional order of uniformity removes one order from that remainder, because the state and the tracial state assign the same value to every power of M below the uniformity order.

Lemma A.10 (Tracial agreement below the uniformity order). Let $|\psi\rangle$ be k -uniform on n qudits. Then $\langle \psi | O | \psi \rangle = q^{-n} \operatorname{Tr} O$ for every operator O supported on at most k parties. In particular, with $M = \sum_j h_j^{(j)}$ as above, $\langle \psi | M^r | \psi \rangle = q^{-n} \operatorname{Tr}(M^r)$ for every $r \leq k$.

Proof. If $O = O_S \otimes I$ with $|S| \leq k$ then $\langle O \rangle = \operatorname{Tr}(\rho_S^\psi O_S) = q^{-|S|} \operatorname{Tr} O_S = q^{-n} \operatorname{Tr} O$. Expanding M^r gives a sum of products of one-site operators touching at most $r \leq k$ sites, so every term is of that shape. $\square$

The tracial state $q^{-n} \operatorname{Tr}$ is the product of the one-site tracial states, and the site summands of M commute, so $q^{-n} \operatorname{Tr} f(M) = \mathbb{E} f(X_1 + \dots + X_n)$ with the X_j independent and X_j uniform on the spectrum of h_j . The tracial main term is therefore a product of one-site characteristic functions, and the following contraction estimate controls each factor.

Lemma A.11 (Per-site contraction). Let h be traceless Hermitian on $\mathbb{C}^q$ with spectral spread $s = \lambda_{\max}(h) - \lambda_{\min}(h)$. Then

$$|q^{-1} \operatorname{Tr} e^{ih}|^2 \leq 1 - c(s) \|h\|_F^2 / q, \quad c(s) = \max \left\{ 1 - \frac{s^2}{12}, \frac{4}{\pi^2} [s \leq \pi] \right\}.$$

Proof. Write $q^{-1} \operatorname{Tr} e^{ih} = \mathbb{E} e^{iX}$ with X uniform on the spectrum. For an independent copy X' , $|\mathbb{E} e^{iX}|^2 = \mathbb{E} \cos(X - X')$ and $|X - X'| \leq s$. From $\cos u \leq 1 - u^2/2 + u^4/24 = 1 - (u^2/2)(1 - u^2/12)$ and $u^2 \leq s^2$, $\mathbb{E} \cos(X - X') \leq 1 - (1 - s^2/12) \operatorname{Var} X$; from $\cos u \leq 1 - 2u^2/\pi^2$ for $|u| \leq \pi$, $\mathbb{E} \cos(X - X') \leq 1 - (4/\pi^2) \operatorname{Var} X$. Tracelessness gives $\operatorname{Var} X = \operatorname{Tr}(h^2)/q = \|h\|_F^2/q$. $\square$

Theorem A.12 (Stability at uniformity order k). Let $|\psi\rangle$ be k -uniform with $k \geq 2$, let each h_j be traceless Hermitian with spectral spread at most s , and keep t, D, U as in Theorem A.8. Put

$$c = c(s), \quad \lambda = \frac{2t^{k-1}}{(k+1)!}, \quad \rho = c - \frac{c^2 D^2}{4q} - 2\lambda.$$

Then $\varepsilon(U)^2 \geq \rho D^2/q$; in particular $\rho > 0$ gives $D \leq \sqrt{q/\rho} \varepsilon(U)$.

Proof. Write $e^{ix} = \sum_{r \leq k} (ix)^r/r! + \varrho(x)$ with $|\varrho(x)| \leq |x|^{k+1}/(k+1)!$. By Lemma A.10 the state and the tracial state agree on every M^r with $r \leq k$, so $\langle e^{iM} \rangle - q^{-n} \text{Tr } e^{iM} = \langle \varrho(M) \rangle - q^{-n} \text{Tr } \varrho(M)$. For any spectral weight vector $p_i \geq 0$ with $\sum_i p_i = 1$,

$$\left| \sum_i p_i \varrho(\lambda_i) \right| \leq \frac{\|M\|_{\text{op}}^{k-1}}{(k+1)!} \sum_i p_i \lambda_i^2,$$

and both states give M^2 the value D^2/q , by Lemma A.10 with $r = 2$ on one side and by tracelessness on the other. Since $\|M\|_{\text{op}} \leq t$, each term is at most $t^{k-1} D^2/(q(k+1)!)$, so the two expectations differ by at most $\lambda D^2/q$. The sites commute, so $q^{-n} \text{Tr } e^{iM} = \prod_j q^{-1} \text{Tr } e^{ih_j}$, and Lemma A.11 with $1 - y \leq e^{-y}$ gives

$$|\langle e^{iM} \rangle| \leq \prod_j (1 - c \|h_j\|_F^2/q)^{1/2} + \lambda D^2/q \leq e^{-cD^2/(2q)} + \lambda D^2/q.$$

With $x = cD^2/(2q)$ and $1 - e^{-x} \geq x - x^2/2$, $\varepsilon(U)^2 = 2 - 2|\langle e^{iM} \rangle| \geq 2x - x^2 - 2\lambda D^2/q = \rho D^2/q$. $\square$

Corollary A.13 (The certified radius grows with the uniformity order). Let $|\psi\rangle$ be k -uniform with $k \geq 2$. If every h_j has spectral spread at most $\frac{1}{2}$, if $D \leq \sqrt{q}/2$, and if

$$t \leq R_k := \left(\frac{(k+1)!}{48} \right)^{1/(k-1)},$$

then $D \leq \sqrt{6q/5} \varepsilon(U)$. One has $\log R_k = \log k - 1 + O(\log k/k)$, so $R_k \sim k/e$.

Proof. Spread at most $\frac{1}{2}$ gives $c \geq 1 - \frac{1}{48} = 0.97917$; with $D^2 \leq q/4$, $c^2 D^2/(4q) \leq c^2/16$, so $c - c^2 D^2/(4q) \geq 0.91924$; and $t^{k-1} \leq (k+1)!/48$ gives $2\lambda \leq \frac{1}{12}$. Hence $\rho \geq 0.83591 > \frac{5}{6}$ and $\sqrt{q/\rho} \leq \sqrt{6q/5}$. The asymptotics are Stirling's formula applied to $\log R_k = (\log(k+1)! - \log 48)/(k-1)$. $\square$

At $k = 2$ this is weaker than Theorem A.8, which reaches $t \leq \frac{1}{2}$ while $R_2 = \frac{1}{8}$; the loss is the passage through the tracial main term, which the direct argument avoids. The gain begins at $k = 3$, where $R_3 = 1/\sqrt{2}$, and grows from there: $R_4 = 1.357$, $R_{10} = 4.548$, $R_{30} = 12.93$. For a stabilizer AME($2m, q$) state $k = m$, so the certified radius grows linearly in the party count. It cannot grow faster.

Proposition A.14 (Ceiling on the certified radius). For every $q \geq 2$ and every n there are traceless Hermitian h_j on $\mathbb{C}^q$ with $\sum_j \|h_j\|_{\text{op}} = 2\pi(1 - 1/q)n$ and $D > 0$ for which $\bigotimes_j e^{ih_j}$ is a scalar, hence $\varepsilon = 0$. No estimate $\varepsilon(U)^2 \geq \rho D^2/q$ with $\rho > 0$ can therefore hold on a region of radius exceeding $2\pi(1 - 1/q)n$.

Proof. Take $h_j = (2\pi/q) \text{diag}(q-1, -1, \dots, -1)$ at every site. It is traceless, and both eigenvalues of e^{ih_j} equal $e^{-2\pi i/q}$, so $e^{ih_j} = e^{-2\pi i/q} I$ and the product of n of them is a global phase. Here $\|h_j\|_{\text{op}} = 2\pi(q-1)/q$ and $\|h_j\|_F^2 = (2\pi/q)^2 q(q-1) > 0$. $\square$

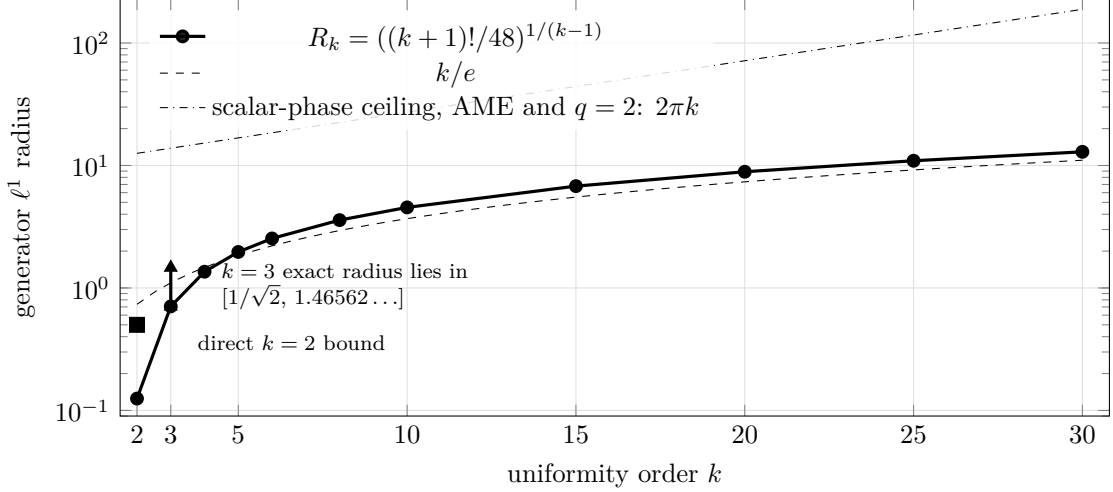


Figure 4: Certified generator ℓ^1 radius of the quadratic stability estimate against uniformity order. The general radius $R_k = ((k+1)!/48)^{1/(k-1)}$ grows as k/e ; scalar-phase configurations bound every such radius by $2\pi(1 - 1/q)n$, so linear growth is the maximal order. For an $\text{AME}(2m, q)$ state, $k = m$. At $k = 3$, the Reed–Muller family pins the exact radius for the constant $\sqrt{6q/5}$ inside the displayed interval.

Linear growth in the party count is therefore the maximal possible order, and Corollary A.13 attains it for AME states, within a factor $4\pi e(1 - 1/q)$ of the ceiling. Bounded uniformity order is a different regime, and the following family measures it exactly.

Proposition A.15 (The certified radius at uniformity order three). Let $\ell \geq 3$, $N = 2^\ell$, and let $|C\rangle$ be the equal-phase CSS state of $C = \text{RM}(1, \ell) \leq \mathbb{F}_2^N$. Its uniformity order is 3, for every ℓ . Take $h_j = cZ(1)$ at every site and put $\theta = Nc \in (0, 2\pi)$. Then

$$\sum_j \|h_j\|_{\text{op}} = \theta, \quad D = \frac{\theta\sqrt{2}}{\sqrt{N}}, \quad \varepsilon = \frac{2|\sin(\theta/2)|}{\sqrt{N}}, \quad \frac{D}{\sqrt{2}\varepsilon} = \frac{\theta}{2\sin(\theta/2)}.$$

The last ratio does not depend on N and increases on $(0, 2\pi)$; it passes $\sqrt{6/5}$ at the unique $\theta^* = 1.46562\dots$. Hence the conclusion of Theorem A.8 fails for every $t > \theta^*$, for every length. At uniformity order three the supremal radius for the constant $\sqrt{6q/5}$ therefore lies between $R_3 = 1/\sqrt{2}$ and θ^* .

Proof. The nonzero weights of $\text{RM}(1, \ell)$ are $N/2$ and N , and $C^\perp = \text{RM}(\ell-2, \ell)$ has minimum distance 4. A marginal of the equal-phase CSS state fails to be maximally mixed only when a nonzero word of C or of $C^\perp$ is supported inside it, so by (2.3) the uniformity order is $\min(N/2, 4) - 1 = 3$ whenever $N \geq 8$. Since $Z(1)|x\rangle = (-1)^x|x\rangle$ at a single site, $U|x\rangle = e^{ic(N-2\text{wt}(x))}|x\rangle$, and the weight distribution 1, $2N-2$, 1 on weights 0, $N/2$, N gives $\langle U \rangle = 1 - (1 - \cos \theta)/N$. This is nonnegative, so $\varepsilon^2 = 2 - 2\langle U \rangle = 4\sin^2(\theta/2)/N$. Also $\|cZ(1)\|_F^2 = 2c^2$ and $\|cZ(1)\|_{\text{op}} = c$, giving the displayed D and θ . Finally $\theta/(2\sin(\theta/2)) = u/\sin u$ with $u = \theta/2 \in (0, \pi)$, which increases from 1 to ∞ ; for every $\theta > \theta^*$ the ratio exceeds the constant of Theorem A.8 while the defect is $2\sin(\theta^*/2)N^{-1/2}$. $\square$

Two consequences. The smallness hypothesis of Theorem A.8 is load-bearing, and its constant $\frac{1}{2}$ is within a factor of about three of sharp, since the family leaves the hypothesis at $\theta = \frac{1}{2}$ and violates the conclusion immediately above θ^* . And the party count is not what governs the family has N parties and uniformity order three at every length, so what it bounds is the radius at order three; the defect at which it breaks the conclusion is $1.3379\dots N^{-1/2}$

only because a bounded ℓ^1 radius spread evenly over N sites has defect image of order $N^{-1/2}$. When the uniformity order grows with the party count, as it does for an AME($2m, q$) state, the admissible generator radius grows with it; the independent constraint $D \leq \sqrt{q}/2$ then binds before $t \leq R_m \sim m/e$ does. This enlarges a region already described in generator coordinates. It does not imply that every unitary in a constant-radius defect ball has such coordinates. Both statements concern a known state acted on by product unitaries of bounded generator norm; neither certifies an unknown device.

A.4 Residual stability from a balanced bipartition

For stabilizer AME states there is a second route to a stability estimate, which buys the same conclusion with a different hypothesis set. It gives up the global ℓ^1 budget on $\sum_j \|h_j\|_{\text{op}}$ entirely, keeps a per-site condition and the global ℓ^2 condition, and pays only in the constant. The mechanism is a bipartition bound in which the Weyl amplitudes of the two halves are compared directly, so no Taylor remainder appears and no budget is needed to control one.

Proposition A.16 (Balanced-bipartition bound). Let $|\psi\rangle$ be a stabilizer AME($2m, q$) state with label group L and character χ , let $U = \bigotimes_j U_j$ be a product unitary, and put $a_j(v) = q^{-1} \text{Tr}(W_v^\dagger U_j)$, so that $p_j = |a_j|^2$ is a probability distribution on $\mathbb{F}_q^2$ for each j . For a set A of m parties define $u, v \in \mathbb{R}^L$ by

$$u(\ell) = \prod_{j \in A} |a_j(\ell_j)|, \quad v(\ell) = \prod_{j \notin A} |a_j(\ell_j)|.$$

Then $\|u\| = \|v\| = 1$ and $|\langle \psi | U | \psi \rangle| \leq \langle u, v \rangle$, so $\varepsilon(U)^2 \geq \|u - v\|^2$.

Proof. The coordinate restriction $\pi_A : L \rightarrow (\mathbb{F}_q^2)^A$ has kernel the labels supported on the complementary m -set, which by (4.1) is trivial; both sides have q^{2m} elements, so π_A is a bijection, and likewise for π_{A^c} . Hence $\|u\|^2 = \sum_{x \in (\mathbb{F}_q^2)^A} \prod_{j \in A} p_j(x_j) = 1$, and the same for v . From (2.3) and $\text{Tr}(W_\ell U) = \prod_j \text{Tr}(W_{\ell_j} U_j)$, taking moduli termwise gives $|\langle \psi | U | \psi \rangle| \leq \sum_{\ell \in L} \prod_j |a_j(\ell_j)| = \langle u, v \rangle$, the relabelling $\ell \mapsto -\ell$ being a bijection of L . For unit vectors $\langle u, v \rangle = 1 - \frac{1}{2} \|u - v\|^2$. $\square$

The AME condition supplies more than the two bijections: it pins down a family of labels of minimum weight that straddles the cut, and those are the labels on which u and v are forced apart.

Lemma A.17 (Cut-transversal labels). With A as above and $j \notin A$, let $L_j = \{\ell \in L : \text{supp } \ell \subseteq A \cup \{j\}\}$. Then $|L_j| = q^2$, every nonzero $\ell \in L_j$ has support exactly $A \cup \{j\}$, and for every $i \in A \cup \{j\}$ the evaluation $\ell \mapsto \ell_i$ is a bijection $L_j \rightarrow \mathbb{F}_q^2$.

Proof. L_j is the kernel of the projection of L to the coordinates $A^c \setminus \{j\}$, which is surjective because it factors the bijection π_{A^c} ; so $|L_j| = q^{2m}/q^{2(m-1)} = q^2$. A nonzero $\ell \in L_j$ has weight at least $m+1$ by (4.1) and support inside a set of that size. For $i \in A \cup \{j\}$ the kernel of $\ell \mapsto \ell_i$ on L_j consists of labels supported on m parties, hence is trivial, and the two sets have the same cardinality. $\square$

Theorem A.18 (Stability under per-site spectral control). Let $|\psi\rangle$ be a stabilizer AME($2m, q$) state with $m \geq 2$, let $U = \bigotimes_j U_j$ be a product unitary, and put $\eta_j = 1 - |q^{-1} \text{Tr } U_j|^2$ and $H = \sum_j \eta_j$. If $H \leq \frac{1}{4}$ then $\varepsilon(U)^2 \geq H/4$. Consequently, if $U = \bigotimes_j e^{ih_j}$ with each h_j traceless Hermitian of spectral spread at most $s \leq \pi$ and $D \leq \sqrt{q}/2$, then

$$\varepsilon(U)^2 \geq \frac{c D^2}{4q}, \quad \text{that is} \quad D \leq 2\sqrt{q/c} \varepsilon(U),$$

with $c = c(s)$ as in Lemma A.11. No bound on $\sum_j \|h_j\|_{\text{op}}$ is assumed.

Proof. The balanced-cut calculation in the proof of Proposition 5.4 establishes $\varepsilon(U)^2 \geq H/4$ before the generator estimates are introduced. Lemma A.11 then gives $H \geq cD^2/q$, while $H \leq D^2/q \leq \frac{1}{4}$, proving the second statement. $\square$

The two routes are a choice of hypothesis set. Corollary A.13 assumes per-site spread $\frac{1}{2}$, $D \leq \sqrt{q}/2$, and the ℓ^1 budget $t \leq R_k$, and delivers the constant $\sqrt{6q/5}$. Theorem A.18 keeps $D \leq \sqrt{q}/2$, relaxes the spread hypothesis to π , deletes the budget, and pays $2\sqrt{q/c}$, which is $\pi\sqrt{q}$ at spread π and $2.02\sqrt{q}$ at spread $\frac{1}{2}$. Its per-site spread condition is not decorative: the family of Proposition A.14 has e^{ih_j} scalar, so every η_j vanishes and the first bound is empty, and those h_j have spread 2π , exactly where $c(s) \leq 0$ and the second bound is empty too. The spread condition is the residue of the ceiling.

One configuration is worth recording, because it is the natural candidate for a product unitary that the balanced-bipartition bound might fail to see. At $q = 2$ take $U_j = H$ at every site, so that each p_j is uniform on $\{X, Z\}$; then $u(\ell)v(\ell) = 2^{-m}$ when every $\ell_j \in \{X, Z\}$ and vanishes otherwise, so $\langle u, v \rangle = 2^{-m} |L \cap \{X, Z\}^{2m}|$. Writing $\{X, Z\} = X + \{0, Y\}$, the set $L \cap \{X, Z\}^{2m}$ is empty or a coset of $L \cap \{0, Y\}^{2m}$, and the latter is a binary linear code of length $2m$ whose nonzero words have weight at least $m + 1$. Singleton bounds its dimension by m . Equality would give a binary $[2m, m, m + 1]$ MDS code. In systematic form $G = [I_m \mid A]$, the MDS property forces every square minor of A to be nonsingular. The 1×1 minors therefore force every entry of A to be 1, whereas for $m \geq 2$ any 2×2 minor is singular. Equality therefore forces $m \leq 1$. Hence $\langle u, v \rangle \leq \frac{1}{2}$ and $\varepsilon(H^{\otimes 2m})^2 \geq 1$ for every stabilizer AME($2m, 2$) state with $m \geq 2$. The bound is not merely nonvacuous there; it is within $\sqrt{2}$ of the largest possible defect. What the balanced-bipartition bound discards is phase information, but the mechanism that detects this configuration is a count of codewords in a structured subset of L , and the MDS constraints force that count down.

A.5 From local stability to exact branch selection

Corollary A.19 (Approximate symmetries decompose). Let $|\psi\rangle$ be a stabilizer AME($2m, q$) state with $m \geq 2$. There is an $\varepsilon_0(\psi) > 0$ such that every product unitary U with $\varepsilon(U) < \varepsilon_0$ factors as

$$U = g \cdot \bigotimes_{j=1}^{2m} e^{ih_j}, \quad g \in G(\psi),$$

with the h_j traceless Hermitian, $\sum_j \|h_j\|_{\text{op}} \leq \frac{1}{2}$, and

$$\left(\sum_j \|h_j\|_F^2 \right)^{1/2} \leq \sqrt{6q/5} \varepsilon(U).$$

By Theorem 1.1 the factor g is an exact local-Clifford symmetry.

The size constraint on the h_j is part of the conclusion and cannot be dropped: the generators of Proposition A.14 are traceless with $\bigotimes_j e^{ih_j}$ scalar, so the defect vanishes while the generator norm does not.

Proof. Work on the compact manifold $X = \text{PU}(q)^{2m}$, on which $f(U) = \varepsilon(U)^2$ is well defined because the defect depends on U only through $|\langle \psi | U | \psi \rangle|$.

Quadratic growth. Let $B \subseteq X$ be the image of $\{(h_j) \text{ traceless} : \sum_j \|h_j\|_{\text{op}} \leq \frac{1}{2}\}$ under $(h_j) \mapsto [\bigotimes_j e^{ih_j}]$. That map is a diffeomorphism near the origin, of matching dimension $2m(q^2 - 1)$, so B is a neighbourhood of the identity coset, and Theorem A.8 gives $f \geq$

$\frac{5}{6q} \sum_j \|h_j\|_F^2$ on it. For $g \in G(\psi)$ one has $f(gV) = f(V)$: if $g|\psi\rangle = e^{i\theta}|\psi\rangle$ then $\langle\psi|gV|\psi\rangle = e^{i\theta}\langle\psi|V|\psi\rangle$, so the modulus is unchanged. Hence the same estimate holds on the left translate gB , with the generators measured from g , which is the form the factorization $U = g \cdot \bigotimes_j e^{ih_j}$ requires.

Isolation and the threshold. The zero set of f on X is the image Z of $G(\psi)$, finite by Theorem A.3 and Remark A.4. So $\bigcup_{g \in Z} gB$ is a neighbourhood of Z , and its complement K is compact. If $K = \emptyset$, then every point of X already lies in some gB , so the desired factorization holds globally and one may choose any $\varepsilon_0 > 0$. Otherwise $f > 0$ on K ; put $\varepsilon_0 = (\min_K f)^{1/2} > 0$. If $\varepsilon(U) < \varepsilon_0$ then $U \notin K$, so $U \in gB$ for some $g \in Z$, and the first step applies at that g . $\square$

The stabilizer hypothesis enters this corollary only in the last sentence of its statement. The factorization itself holds for every 2-uniform state, with g an exact product-unitary symmetry; and finiteness of the zero set is a consequence of (B.1) rather than a separate input, since quadratic growth already isolates the zeros of f on a compact manifold. The threshold this proof produces comes from compactness and is not exhibited; Theorem A.24 below replaces it by a formula, at the cost of the stabilizer hypothesis and of a factor exponential in the party count.

Corollary A.19 measures how far an approximate symmetry of one state is from an exact one. If an exact base intertwiner between two states is available, left translation reduces the two-state problem to this one-state theorem with no loss; this is Corollary 5.6. The proposition below addresses the different problem in which no exact base is assumed: it converts a small global defect into a nearby Clifford at each party, at a threshold exhibited in closed form.

Lemma A.20 (Quantitative diagonal-tensor axes). Let $r \geq 3$, let $E_1, \dots, E_r$ be N -dimensional complex inner product spaces with orthonormal bases $\{e_{ij}\}$, and put

$$T = a \sum_{j=1}^N \lambda_j e_{1j} \otimes \cdots \otimes e_{rj}, \quad a > 0, \quad |\lambda_j| = 1.$$

Let T' have the same form in orthonormal bases $\{e'_{ij}\}$ of spaces E'_i , and let $A_i : E_i \rightarrow E'_i$ be unitaries with

$$\|(A_1 \otimes \cdots \otimes A_r)T - T'\| \leq \eta, \quad \delta = \eta/a < 1/\sqrt{2}.$$

Then in each factor there is a unique permutation π_i with

$$\min_{|\alpha|=1} \|A_i e_{ij} - \alpha e'_{i, \pi_i(j)}\| \leq \sqrt{2} \delta \quad (1 \leq j \leq N). \quad (\text{B.4})$$

Proof. Contract both tensors in the first factor against a target frame vector e'_{1k}^* . The contraction of T' has rank one, and the contraction of $(A_1 \otimes \cdots \otimes A_r)T$ is within η of it. Flattening the latter between its first remaining factor and the last $r-2$ exhibits its singular values as $a|\langle e'_{1k}, A_1 e_{1j} \rangle|$, $1 \leq j \leq N$. By the Eckart–Young theorem the squared mass outside the largest is at most η^2 , hence at most δ^2 after dividing by a . Unitarity of A_1 and completeness of the source basis make the total squared mass one, so the dominant coefficient has square at least $1 - \delta^2 > \frac{1}{2}$. Two orthogonal rows cannot select the same dominant column, and the overlap matrix is square, so the selections form a permutation π_1 . Each matched column also has total squared mass one and contains that coefficient, so its off-axis mass is at most δ^2 , and $2(1 - \sqrt{1 - \delta^2}) \leq 2\delta^2$ gives (B.4). Repeat in every factor. $\square$

Proposition A.21 (Quantitative intertwiner rigidity). Let $q = p^e$, let $m \geq 2$, and let $|\psi\rangle, |\phi\rangle$ be stabilizer $\text{AME}(2m, q)$ states. Put

$$\tau_p = \min\left\{\frac{1}{3}, \frac{\sin(\pi/p)}{2\sqrt{2}}\right\}, \quad \delta = 2q^{(m+1)/2}\varepsilon.$$

Suppose that, after an allowed party permutation,

$$\min_{|\alpha|=1} \|(U_1 \otimes \cdots \otimes U_{2m})|\psi\rangle - \alpha|\phi\rangle\| \leq \varepsilon$$

and that $\delta < \tau_p$. Then for every party i there is a single-qudit Clifford K_i with

$$q^{-1/2} \|U_i - K_i\|_{\text{HS}} \leq \sqrt{2} \delta = 2\sqrt{2} q^{(m+1)/2} \varepsilon. \quad (\text{B.5})$$

For $m = 3$ and equal-phase CSS states of linear $[6, 3, 4]_q$ MDS codes this is the bound $2\sqrt{2} q^2 \varepsilon$ under $2q^2 \varepsilon < \tau_p$.

Proof. Fix an $(m+1)$ -party set S . By (2.3) and Proposition 3.4 the supported label group $L_\psi(S)$ has q^2 elements and projects bijectively onto $\mathbb{F}_q^2$ at every party of S , so the nonidentity part of the reduced operator is

$$T_S = q^{-(m+1)} \sum_{v \neq 0} \lambda_v W_{1, L_1 v} \otimes \cdots \otimes W_{m+1, L_{m+1} v}, \quad |\lambda_v| = 1,$$

where v runs over the nonzero labels at one fixed party of S and each L_i is the resulting parametrization of $L_\psi(S)$ at party i . Normalizing every Weyl operator by $q^{-1/2}$ puts T_S in the form of Lemma A.20 with $r = m+1 \geq 3$, $N = q^2 - 1$, and $a = q^{-(m+1)} q^{(m+1)/2} = q^{-(m+1)/2}$. Two unit vectors at phase-optimized distance ε have density matrices at trace distance at most 2ε ; partial trace contracts the trace norm, and the Hilbert–Schmidt norm is no larger, so every $(m+1)$ -party marginal satisfies $\eta \leq 2\varepsilon$ and hence $\eta/a \leq 2q^{(m+1)/2} \varepsilon = \delta$. The lemma then matches Weyl axes within $\kappa = \sqrt{2} \delta$.

It remains to turn approximate axes into an exact Clifford, party by party. Extend the matched label permutation by $\pi(0) = 0$; the identity axis is matched exactly. Conjugating a product of two Weyl operators in the two available orders places the axes labelled $\pi(v+w)$ and $\pi(v) + \pi(w)$ within 3κ . Distinct Weyl axes are at phase-minimized distance $\sqrt{2}$, so $\delta < \frac{1}{3}$ forces $\pi(v+w) = \pi(v) + \pi(w)$; the matching was a permutation, so π is $\mathbb{F}_p$ -linear. Conjugation preserves Weyl commutators exactly, replacing the four factors of a commutator by matched axes costs at most 4κ , and distinct p -th root commutator phases are separated by $2 \sin(\pi/p)$, so $\delta < \sin(\pi/p)/(2\sqrt{2})$ forces π to preserve the prime-field symplectic form. By (2.1) it is realized by an exact Clifford K_i .

Finally the phases. Put $V = K_i^\dagger U_i$ and expand $V = \sum_x c_x W_x$ with $\sum_x |c_x|^2 = 1$. Axis closeness gives $|q^{-1} \text{Tr}(W_v^\dagger V W_v V^\dagger)| \geq 1 - \kappa^2/2$ for every v , and character orthogonality turns this into $\sum_x |c_x|^4 \geq (1 - \kappa^2/2)^2$. Hence some $|c_x| \geq 1 - \kappa^2/2$, so V lies within κ of a phase times W_x ; absorbing that displacement into K_i gives (B.5). Every party lies in some $(m+1)$ -set, as in Corollary 3.3. $\square$

The exponent in δ is a feature of the states rather than slack in the estimate. The nonidentity part of an $(m+1)$ -party marginal of an $\text{AME}(2m, q)$ state has Hilbert–Schmidt norm $q^{-(m+1)/2} \sqrt{q^2 - 1}$: the marginal is nearly maximally mixed, and the Weyl signal that exposes the local factors is that small, so a global defect is diluted to that scale before any marginal can read it off.

The closest general framework is the perturbation theory of nearly orthogonally decomposable tensors, which gives dimension-independent bounds without a singular-value gap [AY20]; the argument above is the equal-weight complex specialization needed here, together with the Weyl multiplication, commutator, and implementing-Clifford steps. Its conclusion is distance from the ambient Clifford group and nothing finer. Whenever two states admit an exact intertwiner outside a proposed smaller subgroup, that intertwiner has positive distance from the finite closed subgroup, so taking $\varepsilon = 0$ falsifies every uniform estimate toward it.

The exact branch-selection step is Lemma 5.3: on the Clifford locus the defect has no small nonzero values. Thus a nearby product Clifford becomes an exact symmetry without a compactness argument.

A.6 Aggregate minimum-support rounding

The rounding need not inspect one minimum-support marginal in isolation. All such marginals are projections of orthogonal exact-support sectors of the same global density-matrix difference. Summing their squared errors therefore recovers part of the signal lost by marginalization.

Lemma A.22 (Collective minimum-support energy). Let $\rho = |\psi\rangle\langle\psi|$, $\Delta = U\rho U^\dagger - \rho$, $n = 2m$, and $r = m + 1$. Then

$$\sum_{\substack{A \subseteq [n] \\ |A|=r}} q^r \|\Delta_A\|_{\text{HS}}^2 \leq q^n \|\Delta\|_{\text{HS}}^2 \leq 2q^n \varepsilon(U)^2. \quad (\text{B.7})$$

Writing $\delta_A = q^{r/2} \|\Delta_A\|_{\text{HS}}$ and $N_m = \binom{2m}{m+1}$, there are two r -sets A, B with $A \cup B = [n]$ and

$$\delta_A^2 + \delta_B^2 \leq \frac{4q^n}{N_m} \varepsilon(U)^2. \quad (\text{B.8})$$

Proof. The stabilizer expansion of ρ has no nonidentity component on at most m sites. Local conjugation preserves exact support because it preserves tracelessness. Thus Δ_A , for $|A| = r$, is precisely the partial trace of the exact-support- A component of Δ . If that component is $D_A \otimes I_{A^c}$, then $\Delta_A = q^{n-r} D_A$ and

$$\|D_A \otimes I_{A^c}\|_{\text{HS}}^2 = q^{-(n-r)} \|\Delta_A\|_{\text{HS}}^2.$$

Exact-support subspaces are mutually orthogonal, which gives the first inequality in (B.7). The second follows from $\|\Delta\|_{\text{HS}}^2 = 2(1 - |\langle\psi|U|\psi\rangle|^2) \leq 2\varepsilon(U)^2$.

Join two r -sets when their union is $[n]$. This graph is regular: for fixed A , the complement of a neighbor is any $(m-1)$ -subset of A . Averaging $\delta_A^2 + \delta_B^2$ over its edges gives twice the vertex average. One edge therefore satisfies (B.8). $\square$

The two good sets cover every party, so the quantitative-axis argument can be applied to A and B , using A on their intersection. This gives one product Clifford while retaining collective ℓ^2 control of the local errors. Two-uniformity then converts that control directly to state distance, without a telescoping ℓ^1 estimate.

Theorem A.23 (Aggregate global rounding). Let $|\psi\rangle$ be a stabilizer AME($2m, q$) state with $q = p^e$ and $m \geq 3$. Put

$$N_m = \binom{2m}{m+1}, \quad r = m+1, \quad d_p = (2 - 2p^{-1/2})^{1/2}, \quad (\text{B.9})$$

$$R_{m,q} = \min \left\{ \frac{\tau_p \sqrt{N_m}}{2q^m}, \frac{\sqrt{N_m}}{2q^{m+1/2}}, \frac{\sqrt{N_m}}{2\pi\sqrt{2r}q^m}, \frac{d_p}{1 + \pi\sqrt{2rq^{2m}/N_m}} \right\}.$$

If $\varepsilon(U) < R_{m,q}$, then, up to a global phase,

$$U = g \bigotimes_{i=1}^{2m} e^{ih_i}, \quad g \in G(\psi),$$

where the h_i are traceless Hermitian, have spectral spread at most π , and satisfy

$$\left(\sum_i \|h_i\|_F^2 \right)^{1/2} \leq \pi\sqrt{q}\varepsilon(U). \quad (\text{B.10})$$

Moreover $R_{m,q} = \Theta_q(m^{-3/4}(2/q)^m)$. Thus this route is polynomial in m at $q = 2$, whenever the stated AME state exists, improves the exponential base at $q = 3$, improves the frame-rounding prefactor at $q = 4$, and is superseded by Theorem A.24 for $q > 4$.

Proof. Choose A, B from Lemma A.22. The nonidentity part of each marginal has diagonal Weyl coefficient $q^{-r/2}$, so Lemma A.20 and its additivity, commutator, and phase steps give a Clifford K_i at each party with

$$q^{-1/2}\|U_i - K_i\|_{\text{HS}} \leq \sqrt{2}\delta_i, \quad \sum_i \delta_i^2 \leq r(\delta_A^2 + \delta_B^2) \leq \frac{4rq^n}{N_m}\varepsilon(U)^2, \quad (\text{B.11})$$

where $\delta_i = \delta_A$ on A and $\delta_i = \delta_B$ on $A^c \subset B$. The first clause of (B.9) supplies the axis threshold τ_p .

Choose principal eigenangles of $K_i^\dagger U_i$ after optimizing its scalar phase and subtract their mean, obtaining traceless h_i . Since $|x| \leq (\pi/2)|e^{ix} - 1|$ for $|x| \leq \pi$,

$$D^2 := \sum_i \|h_i\|_F^2 \leq \frac{\pi^2 q}{2} \sum_i \delta_i^2 \leq \frac{2\pi^2 q r q^n}{N_m} \varepsilon(U)^2. \quad (\text{B.12})$$

The second clause of (B.9) puts all eigenangles in an arc of length π ; the third gives $D \leq \sqrt{q}/2$.

Put $K = \bigotimes_i K_i$ and $V = K^\dagger U = \bigotimes_i e^{ih_i}$, up to phase. If $M = \sum_i h_i^{(i)}$, two-uniformity and tracelessness give $\langle M^2 \rangle = D^2/q$. Hence

$$\|(V - I)\psi\|^2 = \langle 2 - 2\cos M \rangle \leq \langle M^2 \rangle = D^2/q.$$

Together with (B.12), the fourth clause of (B.9) puts the rays of $K\psi$ and ψ less than d_p apart. Corollary 5.3 forces $K \in G(\psi)$. Apply Theorem A.18 to the residual V : its constant has $c \geq 4/\pi^2$, giving (B.10).

Finally $N_m \sim 4^m/\sqrt{\pi m}$. The third and fourth clauses of (B.9) have order $m^{-3/4}(2/q)^m$, and the other two are larger by a polynomial factor. Comparison with the $q^{-(m+2)/2}/m$ scale below gives the stated boundary at $q = 4$. $\square$

The theorem separates frame recognition from the choice of residual metric. If the ℓ^1 conclusion of Corollary A.19 is required, the same covering pair and a telescoping estimate give it below

$$\min \left\{ \frac{\tau_p \sqrt{N_m}}{2q^m}, \frac{\sqrt{N_m}}{4\pi \sqrt{2q} \sqrt{(m+1)^2 + (m-1)^2} q^m} \right\}. \quad (\text{B.13})$$

At $q = 2$ this is of order $m^{-5/4}$. The loss relative to (B.9) comes from imposing a global generator ℓ^1 budget after the frames are known; it is not a remaining marginal-dilution loss.

A.7 The single-marginal decomposition threshold

Theorem A.24 (Explicit decomposition threshold). Let $|\psi\rangle$ be a stabilizer AME($2m, q$) state with $q = p^e$, $m \geq 2$, $n = 2m$. Put

$$\varepsilon_0 = \min \left\{ \frac{\tau_p}{2q^{(m+1)/2}}, \frac{1}{4\sqrt{2}\pi n q^{(m+2)/2}} \right\}, \quad \tau_p = \min \left\{ \frac{1}{3}, \frac{\sin(\pi/p)}{2\sqrt{2}} \right\}.$$

Every product unitary U with $\varepsilon(U) < \varepsilon_0$ satisfies the conclusion of Corollary A.19.

Proof. Write $\varepsilon = \varepsilon(U)$ and $B = 2\sqrt{2}n q^{(m+2)/2}\varepsilon$, so that $\varepsilon < \varepsilon_0$ gives $B < 1/(2\pi)$.

Rounding. Since $2q^{(m+1)/2}\varepsilon < \tau_p$, Proposition A.21 applied with $|\phi\rangle = |\psi\rangle$ and the identity permutation supplies Cliffords K_i with $\|U_i - K_i\|_{\text{op}} \leq \|U_i - K_i\|_{\text{HS}} \leq 2\sqrt{2}q^{(m+2)/2}\varepsilon$. Summing over the n parties gives $\sum_i \|U_i - K_i\|_{\text{op}} \leq B$.

The rounded Clifford is exact. Put $K = \bigotimes_i K_i$, so that $\|U - K\|_{\text{op}} \leq B$ and $|\langle \psi | K | \psi \rangle| \geq 1 - \varepsilon^2/2 - B$. Here $B < 1/(2\pi) < 0.1592$, and $\varepsilon \leq \tau_p/(2q^{(m+1)/2}) \leq 1/(6 \cdot 2^{3/2}) < 0.06$

because $q \geq 2$ and $m \geq 2$, so the right side exceeds $1 - 0.0018 - 0.1592 > 1/\sqrt{2} \geq p^{-1/2}$. By Lemma 5.3, $K \in G(\psi)$; set $g = K$.

The residual generators are small. Each factor of $V = g^\dagger U$ satisfies $\min_\varphi \|K_i^\dagger U_i - e^{i\varphi} I\|_{\text{op}} \leq \|U_i - K_i\|_{\text{op}} \leq B < 1$. For a unitary W on $\mathbb{C}^q$ with $\sigma = \min_\varphi \|W - e^{i\varphi} I\|_{\text{op}} \leq 1$, the eigenvalues of $e^{-i\varphi} W$ at the optimal phase are $e^{i\mu_a}$ with $2|\sin(\mu_a/2)| \leq \sigma$ and $\mu_a \in (-\pi, \pi]$, so $|\mu_a| \leq 2 \arcsin(\sigma/2) \leq \pi\sigma/2$; subtracting the mean gives $W = e^{i\varphi'} e^{ih}$ with h traceless Hermitian and $\|h\|_{\text{op}} \leq 2 \max_a |\mu_a| \leq \pi\sigma$. Applying this at each site, $\sum_i \|h_i\|_{\text{op}} \leq \pi B < \frac{1}{2}$, and the collected phases are a global phase absorbed into g . Finally $\varepsilon(V) = \varepsilon(U)$, because g fixes the ray of $|\psi\rangle$, so Theorem A.8 applies to V on its hypothesis and gives $D \leq \sqrt{6q/5} \varepsilon(U)$. $\square$

Nothing in this proof is a compactness argument: the finite zero set is never enumerated and no minimum over a compact remainder is taken. What compactness supplied in Corollary A.19 is supplied here by Lemma 5.3, in closed form. The factorization is also constructive in the sense that matters: g is produced from U by rounding each local factor to a Clifford, and an $(m+1)$ -party marginal certifies the rounding.

For this single-marginal route the price is the size of ε_0 , which is $q^{-(n+4)/4}/n$ up to constants and therefore exponentially small in the number of parties. Two steps pay it. Rounding costs $q^{(m+1)/2}$, and by the marginal-dilution remark after Proposition A.21 no detector that reads a single small marginal can avoid that factor. Summing n per-site operator-norm errors and taking n logarithms costs a further $n\sqrt{q}$, which is the price of a per-site conclusion.

The exponential gap left by this theorem and Theorem A.23 is closed by Theorem 5.5. The remaining comparison is with Proposition A.15. That proposition bounds any threshold valid across states of uniformity order three; an $\text{AME}(2m, q)$ state has uniformity order m , and for such states Corollary A.13 certifies the quadratic estimate on a generator-coordinate region whose ℓ^1 radius grows with m , while Theorem A.18 removes that budget for every $m \geq 2$. Those estimates control defect from specified generators; they do not put a defect ball inside that coordinate region. The cleaning theorem supplies the inverse statement at radius of order $n^{-1/2}$, with no claim that this party-count dependence is optimal.

A.8 Local gauge groups of two-unitary gates

Corollary A.25 (Local gauge groups of 2-unitary gates). Let W be a two-qudit gate whose associated four-party state $\psi_{ijkl} = q^{-1} W_{kl,ij}$ is $\text{AME}(4, q)$ — equivalently, W is 2-unitary, remaining unitary after realignment and after partial transposition — and let $G(\psi)$ be its product-unitary symmetry group. Then $(u_1, u_2, u_3, u_4) \mapsto (\bar{u}_1, \bar{u}_2, u_3, u_4)$ is an isometric group isomorphism carrying $G(\psi)$ onto the local gauge group

$$\{(v_1 \otimes v_2, u_3 \otimes u_4) : (u_3 \otimes u_4)W(v_1 \otimes v_2)^\dagger = e^{i\varphi} W\},$$

and it preserves $(\sum_j \|h_j\|_F^2)^{1/2}$. Consequently the local gauge group of any 2-unitary gate is finite modulo the local phases, Theorem A.8 transports with its hypothesis $\sum_j \|h_j\|_{\text{op}} \leq \frac{1}{2}$, and Corollary A.19 transports with its threshold; for a gate arising from a stabilizer $\text{AME}(4, q)$ state Theorem A.24 exhibits that threshold at $m = 2$, namely $\varepsilon_0 = \min\{\tau_p/(2q^{3/2}), 1/(16\sqrt{2}\pi q^2)\}$. Gate defects are measured in the metric $q^{-1}\|\cdot\|_F$. For a gate arising from a stabilizer $\text{AME}(4, q)$ state the exact gauge group is local Clifford.

Proof. Normalization is consistent: $\|W\|_F^2 = q^2$ for a unitary on $\mathbb{C}^q \otimes \mathbb{C}^q$, so $\langle \psi | \psi \rangle = \|W\|_F^2/q^2 = 1$. A product unitary acts on the reshaped gate by $W \mapsto (u_3 \otimes u_4)W(u_1 \otimes u_2)^T$, so $\|U|\psi\rangle - e^{i\theta}|\psi\rangle\| = q^{-1}\|(u_3 \otimes u_4)W(u_1 \otimes u_2)^T - e^{i\theta}W\|_F$; this is the stated metric. Transposition preserves the Frobenius norm and $u \mapsto \bar{u}$ is an automorphism of $U(q)$ fixing $\|h\|_F$ for the corresponding generators, and $(u_1 \otimes u_2)^T = (v_1 \otimes v_2)^\dagger$ with $v_a = \bar{u}_a$, which gives the displayed isomorphism. An $\text{AME}(4, q)$ state is 2-uniform, so Theorem A.3 applies, with Remark A.4 identifying the quotient as the one by the local phases; the stabilizer case is Theorem 1.1. $\square$

The statement is vacuous at $q = 2$, where no $\text{AME}(4, 2)$ state exists. In the dual-unitary circuit literature, local-unitary equivalence is a central part of the classification problem [RAL22], and the finite-field Clifford subfamily has been classified, with perfect-tensor cores appearing among its equivalence classes [Pah26]; the corollary says that for 2-unitary gates the gauge group itself is finite and, in the stabilizer case, Clifford.

B Verification and trust boundary

The principal results are conceptual proofs. This paper contains no paper-facing finite census, generated certificate, or numerical experiment. Its trust boundary has three layers: complete manuscript proofs, selected kernel-checked algebraic cores, and standard results imported by citation.

The formal source namespace is `RelativeConicArcs.AMELU`. In the checked Lean toolchain `v4.32.0-rc1`, the relevant exact cores lie in the support-profile, tensor-rigidity, holonomy-centralizer, stabilizer-dictionary, encoder-transversal, and multipartite modules. The dedicated `AMELUTwoUniformRigidity` gate now audits twenty-two quantitative-core declarations, including the exact-base relative reduction, with only `propext`, `Classical.choice`, and `Quot.sound`. The pre-split aggregate spans a wider development than this paper: it also carries the six-party MDS–CSS applications that appear in the companion paper [Rudd26], so it is not a closure specific to the results here, and no content-addressed contract identifies a subset of it as this paper’s own. Each row of Table 1 therefore states which algebraic core is kernel checked, and the paper claims no end-to-end formalization of any theorem family.

The cleaning theorem uses three cited exact inputs: stabilizer cleaning, localization of nested commutators, and the exact three-cleanable-region Clifford obstruction. The manuscript proves the leakage estimate, the finite Weyl–Fourier concentration step, the stabilizer overlap gap, and their AME composition. Equations (5.3)–(5.11) are closed-form deductions; they do not depend on a computational search or floating-point evaluation.

The quantitative appendix serves as a mechanism comparison. Its single-marginal and aggregate thresholds are weaker than the cleaning radius, and none is used to prove Theorem 1.3.

Table 1: Trust boundary for the theorem families retained in this paper. “Core formalization” means that the named algebraic step is kernel checked; it does not promote the complete manuscript theorem to a formal theorem.

Result family	Manuscript proof	Formal status
AME support counting and complete Weyl-basis rigidity	Sections 2–3	coordinate squeeze, unique local labels, and the abstract axis criterion are kernel checked; the concrete additive-state marginal composition is not
Minimum-support transition maps and phase correction	Section 4	minimum-support generation, the abstract holonomy centralizer, and the stabilizer-character algebra are kernel checked; the state-ray correction is a manuscript step
Encoder Clifford no-go	Section 4	the Choi orientation and Clifford closure are kernel checked; the complete arbitrary-additive encoder theorem is assembled in the manuscript
Cleaning, Weyl–Fourier rounding, branch selection, and residual stability	Section 5	manuscript only; neither the leakage constant 8, the Fourier rounding lemma, the balanced-cut estimate, nor the global radius is formalized
Relative exact-base intertwiner reduction	Section 5	exact line transport, defect invariance, product-intertwiner composition, and lossless transfer of a supplied threshold and coefficient are kernel checked; the cleaning inputs remain manuscript proofs
Robust transition compatibility and character obstruction	Section 5	manuscript only
Two- and k -uniform stability and alternative radii	Appendix A	only the generator splitting, single-exponential identity, polarized second-moment identity, and absence of a nonscalar continuous symmetry are kernel checked; all quantitative constants are manuscript only

AI assistance disclosure

This project was developed with extensive assistance from OpenAI Codex and Anthropic Claude. Under the author’s direction, the systems assisted throughout with proof exploration, literature research, symbolic and finite computation, code and formal-proof development, verification, and manuscript drafting and revision. The author checked the resulting arguments, computations, code, and cited sources, reviewed and edited all AI-assisted material, and assumes responsibility for all content.

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